



**OSTIM TECHNICAL
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**T.C.
OSTIM TECHNICAL UNIVERSITY
Faculty of Engineering**

Department of Electrical and Electronics Engineering

Design and Implementation of a Search and Rescue Robot Dog for Disaster Areas

Graduation Project II

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210202015 Sıla ÖZDEMİR
210202009 Berna Meltem YILDIRIM
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Supervisor

Dr. Şenol GÜLGÖNÜL

May 2026
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GRADUATION PROJECT APPROVAL FORM

The undergraduate graduation project titled “[Afet Bölgelerinde Arama Kurtarma Amaçlı Robot Köpek Tasarımı ve Uygulaması]”, prepared by [Uygar BAŞ, Sıla ÖZDEMİR, Berna Meltem YILDIRIM, Ahmet Emirhan GÖKTÜRK] under the supervision of

[Dr. Şenol GÜLGÖNÜL] was examined and accepted as an Undergraduate Graduation Project in terms of its scope and quality.

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Jury member 2 : Prof. Dr. Sinan KIVRAK

Head of the Department : Prof. Dr. İsmail Hakkı ALTAŞ

FOREWORD

This graduation project was prepared by the students of the Department of Electrical and Electronics Engineering, Faculty of Engineering, OSTİM Technical University, within the scope of Graduation Project II. The project focuses on the design and implementation of a quadruped robot dog intended for search and rescue operations in disaster areas.

We would like to take this opportunity to express our thanks to Asst. Prof. Dr. Senol GULGONUL, our project supervisor and mentor, for the valuable support and guidance that he provided us during the development of our final project. His knowledge and encouragement were critical to successfully completing our graduation project.

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Finally, we would like to express our heartfelt appreciation to all our families for their continuous support, patience, and encouragement throughout our academic journey, including the completion of our final project.

May 2026

Uygar BAŞ
Sıla ÖZDEMİR
Berna Meltem YILDIRIM
Ahmet Emirhan GÖKTÜRK

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ABSTRACT

Search and rescue operations are greatly challenged by natural disasters, such as earthquakes, due to collapsed structures, unstable terrain, and limited access. The traditional rescuer, human, must take many risks and face time constraints when performing rescues in these types of environments. Robotic systems that can safely and effectively operate in abnormal, hazardous, and/or otherwise challenging environments are becoming more desirable in the field.

In this graduation project, the design and implementation of a quadruped search and rescue robot dog for disaster areas is presented. The proposed robot is designed to move effectively on uneven and debris-filled surfaces, where wheeled or tracked robots are insufficient. Owing to its four-legged structure, the robot can maintain balance and mobility in challenging terrains commonly encountered in disaster zones.

The prototype is composed of three functional categories: mechanical systems, electronics, and embedded systems, which are combined into a single integrated prototype. The stability of the actuators allows the robot to maintain a consistent body posture and walk smoothly. A PID-based control approach was used to achieve balanced mobility on uneven ground. The prototype contains an inertial measurement unit (IMU) for orientation feedback and a wide-angle camera for visual perception, which together support both stability control and image-based victim identification. LiDAR and microphone-array integration are planned as future extensions of the system. Sensor readings are processed in real time, so that, except for occasional manual intervention, tracking and detection are handled automatically. Detected signals and their locations are transmitted wirelessly to a remote command-and-control station.

In both cases of being semi-autonomous and fully autonomous, there will be a system with a level of flexibility based on each respective mission. The overall purpose of this project involves increasing the efficiency of search and rescue, improving safety for personnel, and still providing a means for increasing situational awareness in disaster areas. The developed prototype will be a low-cost, easily manufactured robotic solution that supports future research and development in disaster response robotic solutions.

SYMBOLS AND ABBREVIATIONS

AI	Artificial Intelligence
CAD	Computer-Aided Design
CSI	Camera Serial Interface
DOF	Degree of Freedom
IK	Inverse Kinematics
IMU	Inertial Measurement Unit
I2C	Inter-Integrated Circuit
Li-ion	Lithium-ion Battery
PID	Proportional–Integral–Derivative
PWM	Pulse Width Modulation
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver–Transmitter
USB	Universal Serial Bus
W_{total}	Total weight of the robot (N)
W_{leg}	Load/weight per leg (N)
m	Mass (kg)
g	Gravitational acceleration (m/s^2)
F_{leg}	Force acting on one leg (N)
F	Applied force (N)
T, τ	Torque ($N \cdot m$)
r	Lever arm / moment arm (m)
L	Distance from joint axis to load point (m)
σ	Normal stress (Pa)
M	Bending moment ($N \cdot m$)
c	Distance from neutral axis (m)
I_A	Area moment of inertia (m^4)
A	Cross-sectional area (m^2)

x, y Foot position coordinates (m)
 $L1, L2$ Link lengths of the leg mechanism (m)
 $\theta1, \theta2$ Joint angles (rad or deg)
 m_i Mass of component i (kg)
 x_i, y_i Position of component i (m)
 x_{cm}, y_{cm} Center of mass coordinates (m)
 P_{total} Total power consumption (W)
 P_i Power of component i (W)
 E Energy (Wh or J)
 t Time / operating duration (h or s)
 I_{total} Total current consumption (A)
 I_{servo} Current per servo / servo group (A)
 $I_{control}$ Control electronics current (A)
 C Battery capacity (Ah)
 $f_{control}$ Control loop frequency (Hz)
 $T_{control}$ Control loop period (s)

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1.INTRODUCTION

1.1. General Information

1.1.1. Global and Regional Impacts of Natural Disasters and the Need for Search and Rescue

Natural disasters have always had a major impact on human life, societies, and economies. Events such as earthquakes, floods, landslides, and storms can cause serious damage in a very short time. They may lead to loss of life, destruction of infrastructure, displacement of people, and long-term economic problems. Although natural disasters cannot be completely prevented, their effects can be reduced through better planning, preparedness, and effective emergency response systems.

In recent years, the impact of natural disasters has become more visible due to several factors. Climate change, rapid urbanization, population growth, and the expansion of cities into high-risk areas have increased the vulnerability of many regions. According to reports by the United Nations Office for Disaster Risk Reduction (UNDRR), natural disasters have caused trillions of dollars in economic losses over the last two decades. In addition, hundreds of millions of people have been affected through death, injury, displacement, or loss of property.

Türkiye is located in one of the most seismically active regions in the world. The country lies on the Alpine-Himalayan seismic belt and is affected by several major fault lines, including the North Anatolian Fault, the East Anatolian Fault, and the West Anatolian Fault. Because of this geological structure, a large part of Türkiye is under earthquake risk. Past earthquakes such as the 1999 Marmara earthquake, the 2011 Van earthquake, and the Kahramanmaraş earthquakes on February 6, 2023, clearly showed the importance of disaster preparedness and effective search and rescue operations.

The Kahramanmaraş earthquakes affected a very wide area and occurred under difficult winter conditions. These factors made rescue operations more challenging. Collapsed buildings, blocked roads, cold weather, and communication problems created serious difficulties for both rescue teams and survivors. This situation also showed that traditional search and rescue methods may not always be sufficient in large-scale disasters.

One of the most important concepts in search and rescue operations is the “Golden Hours” principle. In general, the first 72 hours after a disaster are considered the most critical period for saving trapped survivors. During this time, the chances of survival are higher. However, after this period, risks such as dehydration, hypothermia, blood loss, and crush syndrome increase significantly. Therefore, reaching survivors quickly and safely is very important.

However, accessing people trapped under rubble is not easy. Collapsed reinforced concrete structures create unstable and dangerous environments. Rescue teams may face aftershocks, falling debris, narrow spaces, and physically exhausting working conditions. As time passes, fatigue and safety risks reduce the effectiveness of human-led rescue operations. For this reason, robotic systems can play an important supporting role in disaster response.

Search and rescue robots are designed to operate in environments that may be too dangerous or difficult for human rescuers. These robots can enter narrow spaces, inspect collapsed structures, collect visual and audio data, and help locate trapped victims. Some systems can also deliver basic supplies or assist rescue teams by providing real-time information. In this way, robotic technologies can improve the speed, safety, and efficiency of search and rescue operations.

1.1.2. Evolution of Mobile Robotic Systems: Transition from Wheeled to Legged Architectures

The development of search and rescue robots is closely related to the development of mobile robotic systems. Early mobile robots were mostly designed with wheels or tracks because these mechanisms are simple, reliable, and easier to control compared to legged systems.

Wheeled robots are efficient and fast on smooth and flat surfaces such as roads, floors, and concrete areas. They also consume less energy and have relatively simple mechanical designs. However, disaster environments are usually not smooth or predictable. After an earthquake, the ground may be covered with broken concrete, metal bars, mud, gravel, collapsed stairs, and other irregular obstacles. In such

conditions, wheeled robots may lose mobility, especially when the obstacle size is larger than the wheel diameter.

Tracked robots were developed to overcome some of these limitations. Since tracked systems have a larger contact area with the ground, they can provide better traction and stability on rough terrain. For example, tracked robots such as PackBot have been used in military and emergency response applications because they can climb stairs and move over uneven surfaces more effectively than many wheeled robots. However, tracked robots also have disadvantages. They are generally heavier, consume more power, and may still become stuck in highly irregular rubble environments.

Legged locomotion provides another solution for moving through complex terrain. In nature, many animals use legs to move across mountains, forests, rocks, and other uneven environments. This idea has inspired the development of legged robots. Unlike wheeled or tracked systems, legged robots do not need a continuous flat surface. They can place their feet on selected stable points, step over obstacles, climb slopes, and adjust their body posture according to the terrain.

Among legged robots, quadruped robots offer a balanced design. Compared to bipedal robots, quadrupeds have better stability because they can keep three legs on the ground while moving the fourth leg. Compared to hexapod or octopod robots, quadrupeds are mechanically simpler and can move faster with fewer actuators. For these reasons, quadruped robots have become an important research area for search and rescue applications.

1.1.3. Dynamics of Quadruped Robots and Control Theory

Designing and controlling a quadruped robot is a challenging engineering problem. Unlike wheeled robots, legged robots require continuous balance control. A wheeled robot can remain stable when it stops, but a quadruped robot must actively control its posture to stay upright, especially while walking on uneven terrain.

The control problem of a quadruped robot can mainly be examined under two topics: kinematics and dynamics. Kinematic modeling deals with the geometric relationship

between the robot body, joints, and foot positions. In other words, it calculates how the movement of each joint affects the position of the robot's legs.

Forward kinematics is used to calculate the position of the foot based on known joint angles. For example, if the angles of the motors in a leg are known, forward kinematics can determine where the foot is in space. Inverse kinematics works in the opposite direction. It calculates the required joint angles for a desired foot position. This is important because the robot must know how to move its joints in order to place its feet correctly during walking.

In a real-time robotic system, these calculations must be performed continuously for each leg. The results are then converted into motor commands. Therefore, both accuracy and speed are important in the control system.

Balance control is another important part of quadruped robot design. The Center of Mass (CoM) of the robot must remain in a stable position relative to the support area formed by the legs touching the ground. When the robot is standing still or moving slowly, stability can be analyzed using the support polygon. However, during more dynamic movements, additional concepts such as the Zero Moment Point (ZMP) may also be considered.

A commonly used method for balance control is the PID controller. A PID controller uses feedback from sensors to reduce the difference between the desired and actual position or orientation of the robot. In this project, an Inertial Measurement Unit (IMU) can be used to measure roll, pitch, and yaw angles. These measurements allow the controller to detect tilting and generate corrective motor commands.

The PID controller consists of three parts:

Proportional (P): Produces a correction based on the current error. If the robot tilts more, the correction becomes stronger.

Integral (I): Corrects accumulated errors over time. This helps reduce small but continuous deviations.

Derivative (D): Responds to the rate of change of error. It helps reduce sudden movements and oscillations.

When the PID parameters are properly tuned, the robot can maintain a more stable body posture while walking or standing on uneven surfaces.

1.1.4. Perception, Mapping, and Artificial Intelligence

A search and rescue robot should not depend completely on manual control, especially in disaster areas. After a major earthquake, communication systems may be damaged. Mobile networks, power lines, and radio communication may not work properly. In addition, concrete structures and underground spaces can weaken wireless signals. Because of these limitations, a useful search and rescue robot should have a certain level of autonomy.

Autonomy requires the robot to sense its environment, process the collected data, and make basic decisions. For this reason, perception and artificial intelligence are important parts of the system. Sensors such as cameras, IMUs, distance sensors, or thermal cameras can help the robot understand its surroundings. The data collected from these sensors can be processed using image processing and machine learning methods.

For example, a camera can be used to detect humans, obstacles, or open paths. Image processing algorithms can help identify important visual features, while artificial intelligence models can support victim detection. Mapping methods can also help the robot create a basic understanding of the environment and support navigation in unknown areas.

In search and rescue applications, perception is not only useful for autonomous movement but also for supporting rescue teams. Visual data, position information, and sensor measurements can help operators make faster and more accurate decisions. Therefore, perception and artificial intelligence increase the practical value of robotic systems in disaster response.

1.1.5. Project Objective and Unique Design Approach

The main objective of this thesis is to design and develop a functional prototype of a low-cost quadruped robot for search and rescue applications. The project aims to create a robotic platform that can move on uneven terrain, collect environmental data, and support disaster response operations.

Commercial quadruped robots, such as Boston Dynamics' Spot, are highly advanced and capable systems. However, their high cost makes them difficult to access for many universities, research groups, and emergency response organizations. For this reason, this project focuses on a more affordable and repairable design approach.

The mechanical structure of the robot is planned to be produced using 3D printing technology. This method provides several advantages. First, it reduces manufacturing cost. Second, it allows damaged parts to be replaced more easily. If a mechanical component breaks during testing or field use, the part can be redesigned, printed, and replaced without the need for expensive manufacturing processes.

The design approach of the project is based on two main principles:

Modular and repairable mechanical structure: The robot is designed with separate leg segments and body parts. This makes maintenance easier and allows damaged components to be changed quickly. The use of 3D-printed parts also supports rapid prototyping and design improvement.

Multi-sensor: The robot uses a 9-axis IMU for orientation measurement and a wide-angle NoIR camera for visual perception. These sensors are processed using a Raspberry Pi 5. The IMU provides information about the robot's roll, pitch, and yaw angles, while the camera provides visual data for environmental observation and possible victim detection. By combining these sensors, the robot can operate more effectively on uneven terrain.

Overall, this project combines mechanical design, electronics, embedded systems, control theory, and artificial intelligence. The expected outcome is a low-cost quadruped robot prototype that can contribute to search and rescue studies. Although the system is developed as a prototype, it aims to demonstrate how affordable robotic platforms can be used to support disaster response operations in challenging environments.

1.2. Literature Review

The Literature Review section analyzes the current state of national and international research regarding Search and Rescue Robots (SRR), Quadruped Walking Mechanisms (QWM), Balance Control Algorithms (BCA), and Sensor Fusion (SF). The analysis

includes both chronological and thematic approaches to the existing SRRs, QWMs, BCAs, and SFs in order. The Literature Review establishes the scientific foundation for this project, highlighting how the proposed SRR is different from those previously researched.

1.2.1. Development of Quadruped Robots and Academic Studies

The earliest studies on quadruped robots can be traced back to the 1980s, when Marc Raibert founded the MIT Leg Laboratory and carried out pioneering research on dynamic legged locomotion. Before modern quadruped designs appeared, single-legged hopping robots were used to study balance and dynamic motion; these platforms maintained stability through continuous hopping under active control [1]. The principles developed in this early work later formed the basis of modern legged robotic systems.

Biswal and Mohanty [2] presented a detailed review of quadruped robots developed over the last two decades. Their study compared several design aspects, including kinematic structure, leg configuration, actuation method, and control strategy. Both mammalian and reptilian leg structures were examined, together with electric, hydraulic, and pneumatic actuation systems. According to the authors, electric motors are generally more suitable for small and medium-sized quadruped robots, since they are more energy-efficient, lighter, and easier to control. Hydraulic actuators, on the other hand, can deliver higher force and load-carrying capacity, which makes them suitable for larger platforms such as Boston Dynamics' BigDog. Since this project focuses on a portable and low-cost quadruped robot for search and rescue applications, electric DS3235 servo motors are selected in line with this recommendation.

MIT's Mini Cheetah is one of the most widely referenced platforms in modern quadruped research. Katz, Di Carlo and Kim [3] reported experimental tests showing dynamic behaviours such as trotting at high speeds, recovery after falling, and 360° backflips. The main innovation of the platform is the use of custom quasi-direct-drive actuators, which provide high torque density with a low gear ratio and allow a more compliant and responsive leg motion.

The same study also showed that quadruped robot control should not rely only on position control; torque and force control are equally important for stable and dynamic locomotion. In this project, standard servo motors are used due to budget limitations. To compensate for the missing torque feedback, a virtual spring-damper approach is planned

in the software so that the prototype can react to ground contact more flexibly. This method cannot match the performance of torque-controlled systems, but it is a reasonable trade-off for a low-cost student platform.

1.2.2. Balance Control Algorithms and Gait Planning

Maintaining dynamic balance and generating stable gait patterns are among the most important challenges in quadruped robotics. A legged robot must continuously control its body posture while moving, especially on uneven or unpredictable terrain. Therefore, balance control and gait planning are essential parts of quadruped robot design.

One of the important theoretical concepts in legged robot balance is the Zero Moment Point (ZMP), introduced by Vukobratović and Borovac [4]. The ZMP defines the point on the ground where the total moment of inertial and gravitational forces becomes zero. As long as the ZMP remains inside the support polygon formed by the contact feet, the robot is considered dynamically stable. Although this concept is more commonly used in humanoid robotics, it also provides a useful theoretical background for legged systems in general.

Quadruped robots usually use different gait patterns depending on the required speed, terrain type, and stability level. Common gait types include walk, trot, pace, bound, and gallop. Each gait has a different leg movement sequence and timing relationship between the legs. For low-speed movement on irregular terrain, walking and trotting gaits are commonly preferred because they provide better stability and control.

Regarding the control algorithm, PID controllers remain one of the most widely used methods in low-cost quadruped platforms, mainly because they are simple, reliable, and computationally lightweight. Earlier studies on IMU-based PID stabilization in other legged and aerial platforms, such as the work of Zul Azfar and Hazry [5], have shown that real-time roll and pitch feedback from an inertial measurement unit, combined with appropriately tuned PID gains, can effectively compensate for orientation disturbances. The same principle can be applied to legged robots, where the body posture is continuously corrected based on IMU data during locomotion.

In this project, a PID-based balance controller driven by IMU feedback is implemented on the Raspberry Pi 5. Gait trajectories are generated through inverse kinematics and executed using the PCA9685 PWM driver. This approach is selected as a

practical trade-off. It does not provide the same dynamic agility as advanced torque-controlled platforms or complex methods such as Model Predictive Control (MPC) and Whole-Body Control (WBC). However, it offers a stable and computationally simple solution that is suitable for the cost-effective hardware approach of the project. For a search and rescue application, stability and reliability are more important than high-speed dynamic motion.

1.2.3. Low-Cost and Open-Source Robot Projects

In recent years, several open-source quadruped projects have shown that legged locomotion is no longer limited to high-budget research laboratories. Notable examples include Stanford Pupper [6], MIT Mini Pupper, and Peto Bittle. These platforms generally use hobby-grade servo motors, 3D-printed structural components, and single-board computers such as Raspberry Pi.

These projects have demonstrated that functional walking platforms can be developed at a much lower cost than industrial robotic systems, which makes quadruped robotics more accessible to students, researchers, and educators. Although these robots have limitations compared to professional platforms, they provide useful examples for learning, prototyping, and testing robotic locomotion.

The present project builds on this open-source philosophy by adopting a similarly cost-effective hardware structure, but extends the approach toward a search and rescue application. Instead of focusing only on walking capability, the proposed system also includes sensor integration, image processing, and basic victim detection functions. In this way, the project combines affordable mechanical design with practical perception features.

1.2.4. Life Detection via Image Processing

Image-based human detection has become an important component of modern search and rescue robotics. This development is mainly supported by recent advances in convolutional neural networks (CNNs) and real-time object detection architectures. Models such as YOLO, introduced by Redmon et al. [7], and later versions such as YOLOv5 and YOLOv8 have shown that human detection can be performed at frame rates suitable for embedded platforms.

Lightweight architectures such as MobileNet SSD, introduced by Howard et al. [8], have also made on-device inference more practical for resource-limited hardware such as Raspberry Pi. This is especially important for low-cost robotic systems, where processing power and energy consumption must be considered carefully.

However, victim detection in disaster scenarios is more difficult than standard human detection. Victims may be partially covered by debris, located in low-light environments, or positioned in unusual body postures. In addition, camera vibration and motion blur caused by robot movement can reduce detection accuracy. Recent studies have addressed these challenges by combining thermal imaging with deep learning models such as YOLO, achieving reliable victim detection even under poor lighting and obstructed conditions [9].

In this project, a YOLO-based detection model running on the Raspberry Pi 5 is used to identify potential victims through the live camera feed. The detected results are then verified remotely by the operator through the FPV interface. This approach combines the speed of automatic detection with the reliability of human judgement, which is especially important in life-critical search and rescue situations.

1.2.5. The Project's Place and Distinctiveness in Literature

As a result of the literature review, it was observed that existing industrial robots such as Spot and ANYmal are highly capable but very expensive, that academic platforms such as the Mini Cheetah [3] are generally developed for laboratory research and experimental control studies, and that low-cost open-source robots are more accessible but usually lack autonomy and artificial intelligence features.

The goal of the study outlined in this thesis is to address the gap in the literature related to cost-effective hardware for search and rescue robotics. The proposed project is distinguished from similar studies by combining a simple and affordable 3D-printed mechanical structure with sensor integration and deep learning-based image processing.

Unlike many low-cost quadruped projects that mainly focus on locomotion, this project also aims to support victim detection and environmental observation. By using affordable components such as the DS3235 servo motors, the Raspberry Pi 5, the HW-579 IMU sensor, and the Pi Camera Module 3, the system aims to provide a functional prototype suitable for search and rescue research. Although the robot is not intended to

compete with industrial platforms, it demonstrates how a low-cost quadruped robot can be adapted for disaster response applications.

1.3. Novelty

The main novelty of this project is the development of a low-cost and modular quadruped robot prototype using accessible components and 3D-printed mechanical parts. Many advanced quadruped robots used in industry are expensive and difficult to reproduce in a student project environment. On the other hand, very simple hobby platforms are usually limited in terms of mechanical strength, sensing capability, and control flexibility. This project aims to build a practical prototype between these two levels.

The first contribution is the use of 3D printing for the robot body and leg parts. The femur, tibia, and chassis-related components are designed to be manufactured with a standard FDM printer. This makes the mechanical structure easier to modify, repair, and reproduce during the development process. For a prototype robot, this is useful because broken or weak parts can be redesigned and printed again without depending on custom manufacturing.

The second contribution is the integration of basic sensing and feedback into the robot platform. The Raspberry Pi Camera Module 3 is used to provide visual information from the robot's surroundings, while the 9-axis IMU is used to observe body orientation. The IMU data is especially important for monitoring pitch, roll, and yaw during motion. In this way, the robot is not only a servo-driven mechanical platform, but also a system that can collect feedback from its own motion and environment.

The third contribution is the use of a simple and expandable control structure. The Raspberry Pi is used as the main control unit, and the PCA9685 driver is used to generate PWM signals for the servo motors. Since low-cost servo motors may have limitations such as backlash, vibration, and torque variation, IMU feedback and PID-based control are considered to improve the stability of the robot in future gait tests. This structure provides a suitable starting point for further development of walking control, balance improvement, and perception-based operation.

Overall, the project does not claim to be a fully autonomous industrial search-and-rescue robot. Instead, its novelty comes from building a functional, low-cost, and

expandable quadruped robot prototype that combines 3D-printed mechanics, servo-based actuation, Raspberry Pi control, camera input, and IMU feedback in a single platform.

1.4. Method

A systems engineering methodology has been utilized to manage and carry out this project as defined by the systems engineering V-Model. The V-Model is a hierarchical system engineering process used to document/ define a project through the various phases of requirements definition, design and implementation, testing, and verification. The project will consist of four major project phases as follows:

Phase 1: Mechanical Design and Kinematic Analysis

Modeling: SolidWorks 3D CAD software was used to create the body and limb components of the robot. In developing the design of the robot, we incorporated the Center of Mass (CoM), which has to be located low to avoid tipping over when walking over obstacles/debris, and included an increase in the Range of Motion (RoM) for the robot's legs to allow for the ability to climb stairs.

Analysis: Using ANSYS's Finite Element Analysis (FEA), the durability of the parts was determined using stress concentrations that were assessed at the leg joints. The results of the analysis were used to optimize the infill density for 3D printed parts.

Kinematics: The inverse kinematics model for the robot was developed using a series of mathematical calculations to obtain a matrix equation that would calculate how many degrees each joint motor needed to rotate for the robot's body to reach the desired position (x, y, z).

Phase 2: Electronic Hardware Integration

Power System: The robot will be powered through the use of several lithium polymer (Li- Po) batteries which provide very high current capabilities. The mechanical drive train/battery pack is powered by a low voltage rating whereas the more sensitive electronic components will be powered through voltage regulation devices (buck regulators).

Communication: Sensors use I2C, microcontrollers communicate with UART, and cameras through USB protocols. Telecommunications data will also be sent from the

robotics to the ground station using the 5 GHz IEEE 802.11ac Wi-Fi link via TP-Link Archer T3U Plus adapters.

Phase 3: Software and Algorithm Development

Robot Operating System (ROS): The software framework of the robot is developed using the Robot Operating System (ROS). ROS is selected because it provides a modular structure in which sensors, actuators, and control units can operate as separate nodes and communicate over a common interface. This makes the software easier to develop, test, and update during the project, and it also keeps the system open to future additions such as LiDAR or mapping modules without changing the rest of the architecture.

Image Processing: The image processing part of the system is developed using the Python programming language and the OpenCV library. For human detection, pre-trained models such as YOLOv5 and MobileNet SSD are considered. These models are optimized for embedded operation and executed on the Raspberry Pi 5. The aim of this part is to detect possible victims from the live camera feed and support the operator during search and rescue operations.

Balance Control: The balance control system uses Euler angles (roll, pitch, and yaw) obtained from the IMU sensor. These orientation values are processed in real time, and a PID feedback loop generates the corrective motor commands required to reduce body tilt during locomotion. This approach is preferred because it is simple, lightweight, and suitable for the hardware limitations of the prototype, while still providing sufficient stability for the search and rescue use case.

Phase 4: Test and Verification

Simulation: Before manufacturing the physical prototype, the robot design is tested in a Gazebo simulation environment. The simulation allows the gait algorithms and basic movement behaviour to be evaluated under controlled conditions, so that mechanical or software-related problems can be identified before physical testing begins.

Field Tests: After the first prototype is manufactured, the robot is tested on a simulated debris track in a laboratory environment. This test area includes inclined ramps, rocky surfaces, stairs, and uneven ground conditions to represent the kinds of obstacles expected in real disaster environments. The main performance metrics evaluated during these tests are the distance travelled before losing balance, battery

runtime, image transmission latency, and human detection accuracy. The results obtained from these tests are then used to identify the strengths and limitations of the prototype and to guide improvements in the next design iteration.

1.5. Impact

Upon successful implementation of the project, the anticipated technical, economic, and social deliverables will be as follows:

1.5.1. Social and National Security Impact

Türkiye is located in an earthquake-prone region, and search-and-rescue technology has become a clear necessity after recent events such as the 2023 Kahramanmaraş earthquakes. The robotic platform developed in this project is intended to be deployed in damaged buildings, partially collapsed structures, and tunnels with possible gas leaks before rescue personnel enters the area. This use is often described as "vanguard reconnaissance" in the search-and-rescue literature.

Sending a robot into the area first is expected to provide three main benefits. The safety of the rescue personnel is improved, since the most hazardous parts of the site are observed remotely before any human entry. The probability of reaching victims trapped under debris within the critical first 72 hours is increased, because the robot can scan multiple areas in parallel with the rescue teams. And the overall response time is reduced, since the teams can focus on the locations where the robot has already detected possible signs of life.

1.5.2. Economic and Commercial Impact

Currently, search-and-rescue robots (produced by foreign manufacturers) used by AFAD [10] (the Turkish Emergency Services) and/or Fire Brigade units/Authorities consume a substantial amount of foreign currency; this project will enable Türkiye to manufacture its search-and-rescue robots using local resources at significantly lower costs, thereby improving Türkiye's economy through the reduction of reliance on foreign products. The prototype developed through this project has excellent commercial potential and may be used in a variety of civilian and non-civilian applications, including the Defense Industry, the Mining Sector, and/or the Industrial Inspection sector.

1.5.3. Academic and Scientific Impact

The information gained from the project (i.e., performance of sensor fusion and strength tests on 3D printed parts) will add to the body of knowledge within the academic area of Mechatronics and Robotics. The findings will be reported in the form of papers presented at both domestic and foreign conferences and will also provide an example of "applied robotics" for other students attending the university. In addition, the Embedded System, Artificial Intelligence, and Mechanical Design skills gained through this project will assist the country in meeting its demands for qualified engineers.

1.5.4. Technical and Engineering Impact

In addition to its social, economic, and academic contributions, the project also provides an important technical impact through the integration of mechanical motion, embedded control, and electronic drive systems in a single robotic platform. The robot's motion control structure is based on a Raspberry Pi 5 and a PCA9685 16-channel PWM servo driver, which allows multiple servo motors to be controlled in a stable and organized way. This structure supports the development of a reliable interface between software-generated motion commands and the physical movement of robot joints.

The use of PWM-based servo control contributes to the precise positioning of the robot legs and provides a practical basis for improving locomotion algorithms. In addition, separating the servo motor power line from the logic supply is an important engineering decision, since it helps reduce voltage drops and electrical noise that may occur during high-current motor movements. Therefore, this design approach improves the reliability of the control system and supports the team's experience in motor control, power distribution, and system integration.

1.5.5 Wireless Data Communications

The proposed system uses wireless communication to maintain a continuous link between the robot and the operator-side workstation. Two TP-Link Archer T3U Plus USB Wi-Fi adapters are used in the system: one is connected to the Raspberry Pi 5 on the robot, and the other is connected to the operator's computer. Together, they form the wireless link through which control commands, telemetry, and live video are exchanged.

The selected adapters support the IEEE 802.11ac standard (Wi-Fi 5) on both the 2.4 GHz and 5 GHz bands. The 5 GHz band is preferred for active operation because it offers

higher throughput and lower interference than the 2.4 GHz band, which is important when the robot operates in environments crowded with other electronic equipment. Higher throughput is required because the wireless link must simultaneously carry three classes of data: low-latency control commands sent by the operator, telemetry data such as orientation and battery status sent by the robot, and a live video stream from the Raspberry Pi Camera Module 3 used by the operator for first-person-view monitoring.

The use of two USB Wi-Fi adapters with external high-gain antennas, instead of internal-antenna alternatives, improves signal quality in environments with physical obstacles such as walls and debris between the robot and the access point. This wireless architecture supports remote monitoring, command transmission, and debugging without requiring a fixed wired connection, which is a structural requirement of a robot intended to operate in disaster zones.

1.6. Standards

The International/Electricity standards provide an overview's Reliability, Safety and Conformity through Design and Build of a Robotic Society.

ISO 13482 (Robots and robotic devices — Safety requirements for personal care robots) [11]: The design of a robot must include Safety Procedures (Collision Avoidance) to reduce loss of life while the robot responds to requirements to Evacuate Victims safely.

IEC 62133 (Secondary cells and batteries containing alkaline or other non-acid electrolytes) [12]: Battery Management Systems will ensure that Lithium-Ion Batteries are protected during Charge/Discharge cycles and against Overheating and Short Circuits.

IEEE 802.11 (Wireless LAN Standards) [13]: A robot must have Wireless (data & image) communication with Ground Control Stations in accordance with the Wireless Radio Frequency (2.4Ghz, 5Ghz) specifications.

IPC-2221 (Generic Standard on Printed Board Design): Electronic Circuit Board Design, Power Distribution Board Design, etc., is done according to IPC or other applicable IC Standards for Electrical/Signal Integrity.

ROS REP (Robot Operating System Enhancement Proposals): During Software Development, the ROS Community Naming/Package Structures (REP-103, REP-105) will be used for Code Reusability and Modularity.

Ethical Standards: The principles of personal data privacy (GDPR/KVKK) and ethical guidelines have been upheld while using image processing technology during the course of this project. In addition, it has been accepted that any images collected will only be processed for use in search and rescue operations.

1.7. Work Schedule

The path to be followed in the Engineering Design and Graduation Project process is given in **Table 1.1** as a work-time graph.

In **Table 1.1**, the project is divided into work packages, and the work done in each package and the time intervals in which these series must be completed are specified. To avoid disruption of the work, the time period that must be done is given in the work package. In case any problems that may arise, plans B were considered and the risk analysis in **Table 1.2** was carried out.

Table 1.1. Work-Time Table

WP No	Name and Purpose of the Work Package	Working Team	Time interval	Contribution to the Project
1	Completion of the literature review and determination of unique design objectives for the project.	Ahmet Emirhan GÖKTÜRK (L) Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ	01/10 /2025 - 12 /11 /2025	Conducting literature review, analyzing similar search and rescue robots, and determining system requirements.
2	Completed 3D model of the chassis and leg structure.	Uygar BAŞ (L) Sıla ÖZDEMİR Berna Meltem YILDIRIM Ahmet Emirhan GÖKTÜRK	01 /10 /2025 - 01 /12 /2025	Modeling the mechanical design in a CAD environment and performing center of gravity and stability analyses.
3	Completion of circuit schematics and component selection.	Sıla ÖZDEMİR (L) Ahmet Emirhan GÖKTÜRK Berna Meltem YILDIRIM	01 /12 /2025 - 01 /02 /2026	Creating the integration plan for electronic components (motor driver, sensors, microcontroller, Wi-Fi adapter) and drawing circuit schematics.
4	Completion of the mechanical prototype.	Uygar BAŞ (L) Berna Meltem YILDIRIM Sıla ÖZDEMİR Ahmet Emirhan GÖKTÜRK	01 /12 /2025 - 01 /02 /2026	Manufacturing mechanical parts via 3D printing and assembling robot chassis.
5	Determination of PID parameters and observation of stable motion in simulations.	Sıla ÖZDEMİR(L) Berna Meltem YILDIRIM Uygar BAŞ	01 /01 /2026 - 01 /04 /2026	Developing the PID-based balance control algorithm and conducting simulation tests.

Table 1.1. Work-Time Table (Continuing)

WP No	Name and Purpose of the Work Package	Working Team	Time interval	Contribution to the Project
6	Achieving simultaneous processing of sensor data.	Uygar BAŞ (L) Berna Meltem YILDIRIM Sıla ÖZDEMİR Ahmet Emirhan GÖKTÜRK	01 /03 /2026 - 01 /05 /2026	Integration of sensor systems and performing image processing tests.
7	Establishment of the robot's basic locomotion (movement) capabilities.	Berna Meltem YILDIRIM (L) Sıla ÖZDEMİR Ahmet Emirhan GÖKTÜRK Uygar BAŞ	01 /02 /2026 - 01 /05 /2026	Setup of the embedded system, software integration, and conducting locomotion (walking) tests.
8	Ensuring low-latency (lag-free) data transmission.	Ahmet Emirhan GÖKTÜRK (L) Berna Meltem YILDIRIM Sıla ÖZDEMİR	01 /03 /2026 - 01 /05 /2026	Data transmission tests via Wi-Fi, and optimization of wireless communication.
9	Stable operation of the robot in semi-autonomous and autonomous modes.	Berna Meltem YILDIRIM (L) Uygar BAŞ Sıla ÖZDEMİR Ahmet Emirhan GÖKTÜRK	01 /04 /2026 - 01 /05 /2026	System testing: gait stability, victim detection via camera, and verification of teleoperated movement.
10	Completion of the final report and prototype presentation	Berna Meltem YILDIRIM (L) Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /04 /2026 - 21 /05 /2026	Preparing the final report, evaluating project outputs, and preparing for the presentation.

1.8. Risk Analysis

Table 1.2. Risk analysis and plan B

WP No	Contribution to the Project	Risk Analysis
1	Instability in walking (locomotion) due to weight imbalance in the mechanical design.	PID control parameters will be retuned; mass distribution will be optimized through simulations, and the leg design will be updated. If necessary, the balance sensor (IMU) calibration will be repeated.
2	Delays in the procurement of electronic components (motor drivers, sensors, Wi-Fi adapters, etc.).	Components with identical technical specifications will be sourced from alternative local or online suppliers. Equivalent parts will be used for temporary testing when necessary.
3	Interruption of the Wi-Fi connection or signal attenuation in a disaster environment.	In case of connection loss, the robot will temporarily store data in local memory and transmit it once the connection is re-established. Alternatively, short-range backup communication will be provided via Wi-Fi or LoRa modules.
4	Power insufficiency or battery failure during prototype testing.	A lithium battery protection circuit will be used, and operations will be supported by an external power supply during the testing process. The battery level will be continuously monitored by the software.

1.9. Organization of Work packages

For the project to progress in a planned and timely manner, a different person was designated as the leader in each work package respectively. Each student carried out the work package with discipline. Table 1.3 shows the work packages that each student led and the number of work packages they were involved in.

Table 1.3. Distribution of Work Packages.

Work Package Leader	Leding Work Package	Number of Work Packages
Sıla ÖZDEMİR	3,5	10
Uygar BAŞ	2,4,6	10
Ahmet Emirhan GÖKTÜRK	1,8	10
Berna Meltem YILDIRIM	7,9,10	10

2. THEORETICAL BACKGROUND

2.1. General Information

The theoretical background of this project is based on the main engineering concepts required for the development of a quadruped robotic platform. Since the robot is designed to move on uneven surfaces, the study includes mechanical stability, legged locomotion, sensor feedback, embedded control, communication, and power management topics.

The main control structure of the robot is built around a Raspberry Pi-based system. A camera module and an IMU sensor are used to support environmental perception and orientation feedback. Servo motors are driven through a PCA9685 PWM driver to generate coordinated leg motion, while additional task-related components such as the emergency stop button, wireless communication adapter, and power system are integrated into the overall structure.

The mechanical body of the robot is planned to be produced using PLA filament, which provides a lightweight and manufacturable structure for prototyping. In this chapter, only the theoretical principles directly related to the project are explained. Detailed component selections, mechanical dimensions, electronic connections, and cost analysis are presented in the design section.

2.2. Quadruped Robots and Robot Dynamics

A quadruped robot is a legged mobile platform whose locomotion is produced by the coordinated motion of four articulated legs. Unlike wheeled or tracked vehicles, a legged platform interacts with the ground through discrete contact points. Therefore, balance and stability are among the main design considerations in quadruped robot systems.

The stability of a quadruped robot is commonly analyzed using the support polygon concept. The support polygon is the region formed by the feet that are in contact with the ground. If the vertical projection of the robot's center of mass (CoM) remains inside this region, the robot can be considered statically stable. This condition is especially important for slow walking gaits, where at least three feet remain on the ground during motion.

When the robot moves faster or when one or more legs are lifted from the ground, stability becomes more dependent on the timing of leg movements, body posture, and

the distribution of load between the legs. For this reason, coordinated joint motion and proper actuator selection are important for maintaining stable locomotion.

The motion of a quadruped robot can be described using the Newton–Euler equations of motion applied to a multi-body mechanical system. These equations relate joint torques, link inertias, gravitational effects, and angular accelerations. However, solving the complete dynamic model of a four-legged robot in real time may be computationally demanding. Therefore, simplified models are commonly used during the design stage. In this project, each leg is considered as a simplified planar mechanism for design calculations and motion analysis. The related mathematical relations and design calculations are presented in the design section.

The robot proposed in this project has eight actuated joints distributed across four legs, with two degrees of freedom per leg corresponding to the hip and knee joints. This configuration provides sufficient mobility for basic walking motions on uneven terrain while keeping the mechanical structure and computational complexity within the limits of the selected hardware.

2.3. Sensors and Feedback Systems

Mobile robotic systems operate through the continuous interaction between sensing and actuation. Sensors convert physical quantities such as acceleration, angular velocity, and visual information into electrical signals. These signals are processed by the controller and used to update the commands sent to the actuators. This sensor–controller–actuator loop forms the basis of feedback-controlled robotic systems.

In a quadruped robot, sensors can generally be grouped into two categories. Proprioceptive sensors measure the internal state of the robot, such as orientation, motion, and body posture, and are important for balance and gait control. Exteroceptive sensors observe the surrounding environment and provide information for perception-related tasks. In the proposed system, an Inertial Measurement Unit (IMU) is used as the proprioceptive sensor, while a camera module is used as the exteroceptive sensor.

An IMU combines acceleration, angular velocity, and magnetic field measurements in a single module. The accelerometer measures linear acceleration along three axes and can provide a reference for the vertical direction when the robot is stationary. The gyroscope measures angular velocity around each axis and supports the estimation of

rotational motion. These measurements can be combined through sensor fusion methods, such as complementary filtering or Kalman filtering, to estimate the robot's orientation in terms of roll, pitch, and yaw angles. The magnetometer can also provide heading information and help reduce long-term drift around the vertical axis.

The camera module provides visual information about the robot's surroundings. Visual perception is based on converting reflected light from the environment into a two-dimensional image through an image sensor. The use of a wide-angle camera increases the field of view, which is useful in narrow or cluttered environments. Since the selected camera is a NoIR type, it can also be used with near-infrared illumination under low-light conditions.

In the proposed system, the IMU and camera are connected to the Raspberry Pi. The IMU provides orientation-related feedback through a digital communication interface, while the camera supplies visual data through a dedicated camera interface. These sensor inputs support the control algorithms and provide feedback for the robot's motion and environmental awareness.

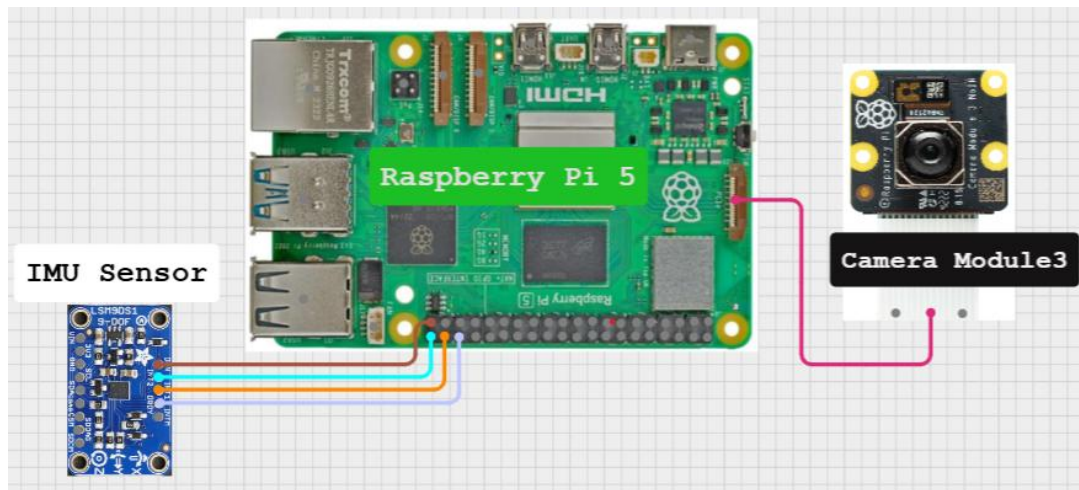


Figure 2.1. Sensor feedback architecture of the robot.

2.4. Wired–Wireless Communication System

Communication in a mobile robotic system can be divided into two main domains depending on the physical medium used to transfer data. Wired communication covers the short-range data exchange between components located inside the robot body, while wireless communication provides the link between the robot and the external operator-side system. These two communication types have different design priorities. Wired

connections are generally preferred for predictable timing, stable signal transfer, and lower sensitivity to external interference, whereas wireless communication provides mobility and remote access.

In the proposed system, on-board data exchange between the central processor and peripheral devices is mainly carried out through wired interfaces. The selected communication method depends on the type and amount of data being transferred. For low-data-rate peripherals, serial bus protocols are commonly used.

The Inter-Integrated Circuit (I2C) protocol is one of these protocols and uses two lines: one for clock and one for bidirectional data. It allows multiple devices to communicate on the same bus using unique addresses. For this reason, I2C is suitable for components such as inertial sensors and motor driver boards.

Other serial communication methods, such as UART and SPI, are also commonly used in embedded robotic systems. UART provides a simple point-to-point communication structure, while SPI offers higher speed at the cost of requiring more signal lines. These protocols are selected according to the communication speed, wiring complexity, and device requirements.

High-speed dedicated interfaces are used when the amount of transferred data is higher. Camera Serial Interface (CSI) is an example of such an interface and provides a direct connection between a camera module and the processor. This type of interface is suitable for visual data transfer because image data requires higher bandwidth than simple sensor measurements. USB is also used for peripherals that connect to the system through standardized ports, such as wireless network adapters.

Wireless communication is used to provide a connection between the robot and the operator-side system without requiring a physical cable. This allows remote monitoring, command transmission, and debugging during system development and operation. The engineering impact of the wireless communication structure is discussed in **Section 1.5.4**.

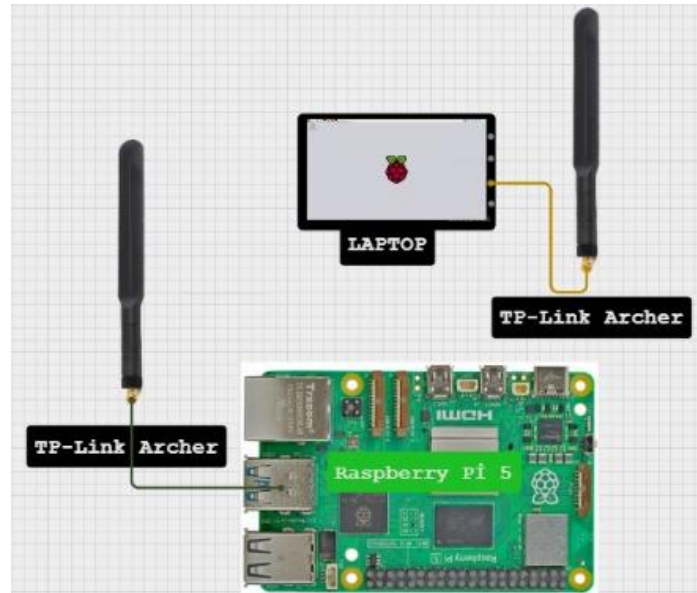


Figure 2.2. Wireless communication architecture of the Raspberry Pi-based robot system.

2.5. Microprocessors and Embedded Control Units

Microprocessors and embedded control units are used in robotic systems to process sensor data, manage communication interfaces, and generate control commands for actuators. In the proposed robot, the Raspberry Pi is used as the main control unit of the system.

The Raspberry Pi receives data from the camera module and IMU sensor and uses this information to support perception, orientation feedback, and motion-related control processes. It also communicates with the PCA9685 servo driver to control the servo motors used in the robot legs.

Different communication interfaces are used according to the requirements of each component. The IMU sensor and PCA9685 servo driver communicate through the I2C protocol, while the camera module is connected through the camera interface. The USB Wi-Fi adapter is used to provide wireless communication between the robot and the operator-side system.

This structure provides a compact and manageable control architecture for the prototype. Since the sensing, communication, and actuator command processes are coordinated through a single main control unit, the electronic system becomes easier to develop, test, and maintain.

2.6. Control Methods

A hierarchical control approach is considered to manage the coordinated movements of the quadruped robot while maintaining balance. The control structure is based on generating suitable joint commands for each leg according to the desired body motion and walking sequence.

In this project, inverse kinematics is used as the theoretical basis for calculating the joint angles required for leg movement. The calculated joint commands are transmitted to the PCA9685 PWM driver through the I2C bus. The driver then generates PWM signals for the servo motors, allowing the robot legs to reach the desired angular positions.

The IMU sensor provides orientation-related feedback such as roll, pitch, and yaw information. This feedback can be used to monitor body tilt during motion and support balance-related control decisions. In this way, the control system combines joint command generation and sensor feedback to support stable locomotion.

2.7. Human-Machine Interaction

Human-machine interaction in a mobile robotic system refers to the communication between the robot and the operator. In this project, the operator-side system is intended to monitor the robot and send basic control commands through the wireless communication link.

The Raspberry Pi Camera Module 3 NoIR provides visual feedback from the robot's surroundings. This camera data can be used by the operator to observe the environment and follow the robot's movement during operation. In addition, the wireless connection supported by the TP-Link Archer T3U Plus USB Wi-Fi adapters allows remote monitoring and command transmission without requiring a physical cable.

This interaction structure improves operational flexibility and makes the robot easier to test, control, and observe during prototype development.

2.8. Power Management and Energy Distribution System

The reliability of a quadruped robot depends strongly on its power management system. The system must provide stable voltage to sensitive electronic components while supplying sufficient current to the servo motors during locomotion.

Energy Source (Li-ion Battery): A 12 V 7500 mAh lithium-ion battery is used as the main energy source of the robot. It supplies power to the Raspberry Pi, camera module, IMU sensor, wireless communication unit, PCA9685 servo driver, and servo motors.

Voltage Regulation: Since the control electronics and servo motors operate at different voltage levels, voltage regulation is required in the power distribution system. Step-down voltage regulation is used to provide suitable voltage levels for the Raspberry Pi and peripheral components.

High-Current Requirement: Servo motors may draw high current during simultaneous leg movements. Therefore, the power system must be able to handle current peaks without causing voltage drops. Stable power delivery is important to prevent unexpected resets of the controller and to maintain reliable servo operation.

Safety and Efficiency: The power distribution structure is arranged to reduce electrical noise, voltage instability, and overheating. An emergency stop button is also included in the system to allow the robot to be shut down safely during testing or operation.

3. DESIGN

3.1. General Information

This section explains the engineering design of the proposed quadruped search and rescue robot developed for operation in disaster areas. The design process is based on mechanical, electrical, and software engineering principles, and aims to create a robust, mobile, and energy-efficient robotic platform capable of navigating uneven terrain such as rubble, debris, and collapsed structures.

A quadruped configuration was selected because of its superior performance (greater stability than wheeled or tracked systems) and adaptability to different surfaces. A quadrupedal robot design draws on existing quadrupedal robot systems as well as an existing open-source reference, but the overall geometry has been modified to fit the requirements of a disaster relief scenario; the dimensions (based on existing designs), materials (based on current technology), actuator placements (based on user specifications), and system integration (based on user requirements) have all been redesigned.

During the design phase, key engineering calculations are carried out to support critical decisions such as:

- Number and placement of servo motors
- Load distribution on each leg
- Required torque for joint actuation
- Power consumption and battery capacity
- Structural strength of 3D-printed components
- Control and communication architecture

Engineering of mechanical components is conducted through the use of Computer Aided Design (CAD) software. Technical drawings will be created; these will include the mechanical component with full numerical dimensioning, as well as the project name, preparation date, designer names, and academic supervision. Electrical design and software design will be created in conjunction with the mechanical design to ensure compatibility with the entire system.

A comprehensive compilation of component parts is compiled at the conclusion of the design phase, along with an initial investigation into costs. Additionally, the legal and regulatory implications regarding the implementation of robotic systems in areas of public interest and disaster response are also conducted.



Figure 3.1. Overall, CAD model of the quadruped robot

3.2. Dimensioning

The physical dimensions of the quadrupedal robots are established in accordance with their ability to successfully navigate restricted or cluttered environments (e.g., earthquake debris) while providing an adequate shield for the electronic components that lie within them. The chassis of each robot is configured as a compact unit, and the control unit, power electronic components, and battery arrangement are integrated so that they will have a location to rest on the robot's body, and, at the same time, will be designed with a low centre of gravity to improve the robot's (overall) stability.

The approximate body dimensions of the robot are planned as follows:

The robot is anticipated to have an overall height from ground to top of robot between approximately 15 cm to 20 cm. Depending on the robots' posture and gait, this height may vary somewhat. In total, the robots' weight- which includes the mechanical structure, electronics, battery, and sensors- is projected to be roughly 5 kg.

The dimensions listed in this section are illustrative only and serve as a reference for the design of the robot and the development of the associated technical drawings. As designs progress through a more detailed CAD model and integration of components further refinements will be made to achieve final dimensional specifications.

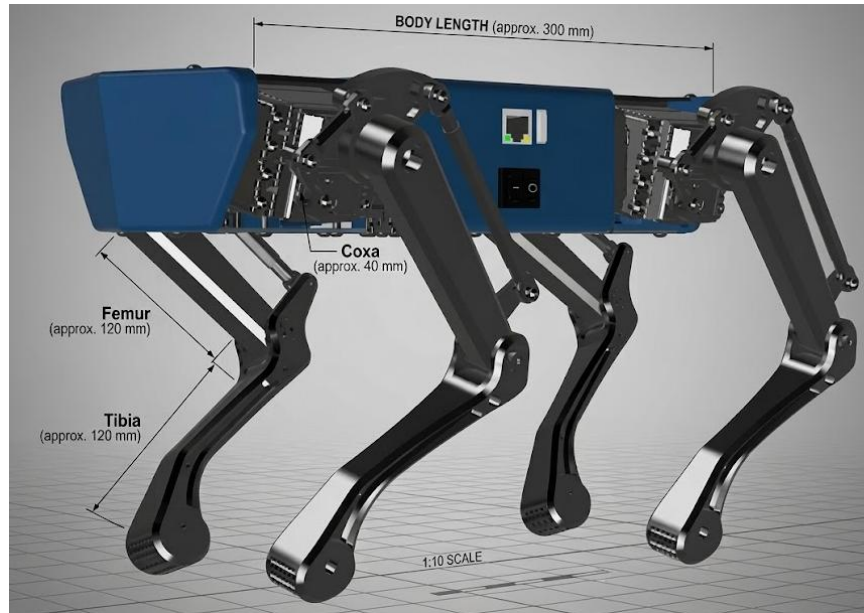


Figure 3.2. Side view of the robot showing overall height and leg geometry.

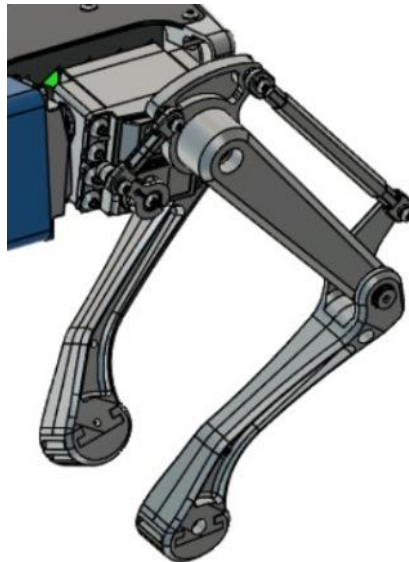


Figure 3.3. Leg segment configuration illustrates coxa, femur, and tibia.

3.3. System Components and Their Selections

The system components are selected according to the mechanical, electronic, control, and communication requirements of the quadruped robot. The proposed system is based on a Raspberry Pi-centered control architecture supported by a camera module, IMU sensor, PCA9685 servo driver, high-torque servo motors, wireless communication adapters, and a Li-ion battery pack.

Each component is selected by considering compatibility with the overall system, power requirements, availability, cost, and suitability for prototype development. The main objective of the component selection process is to create a compact, reliable, and

maintainable robotic platform.

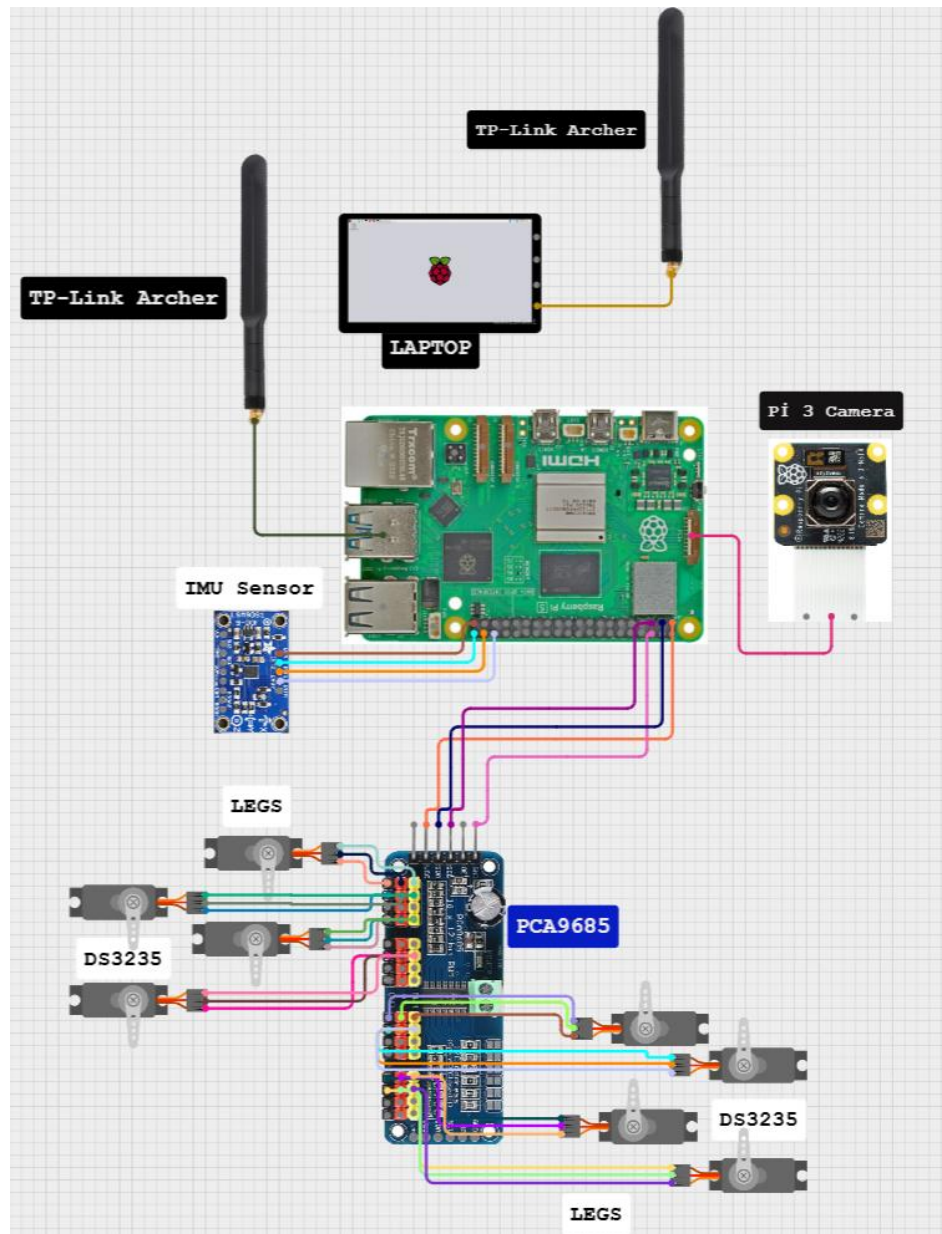


Figure 3.4. Overall electronic connection diagram of the robot.

3.3.1. Raspberry Pi 5

Raspberry Pi 5 was selected as the central processing unit of the quadruped robot system. Owing to its compact structure, low power consumption, and high processing capability, Raspberry Pi 5 is suitable for real-time control applications. Its advanced multi-core ARM processor allows the execution of complex computation tasks such as gait algorithms, balance control, sensor fusion, and image processing.

Through operating system-level multitasking, coordinated control of robot legs, generation of motor drive signals, and processing of sensor data can be managed

efficiently. The processing capacity provided by Raspberry Pi 5 enables stable operation even under demanding computational loads required for quadruped locomotion.

The power requirement of 5 V – 5 A is supplied through an independent regulated power line separated from the main battery system in order to prevent voltage fluctuations from affecting the control electronics. In addition, active cooling modules are integrated to keep processor temperatures within safe operating limits, preventing thermal instability during intensive workloads.

In conclusion, Raspberry Pi 5 is considered an appropriate central controller for the quadruped robot system due to its computing performance, expandability, and suitability for real-time robotic control tasks [14].

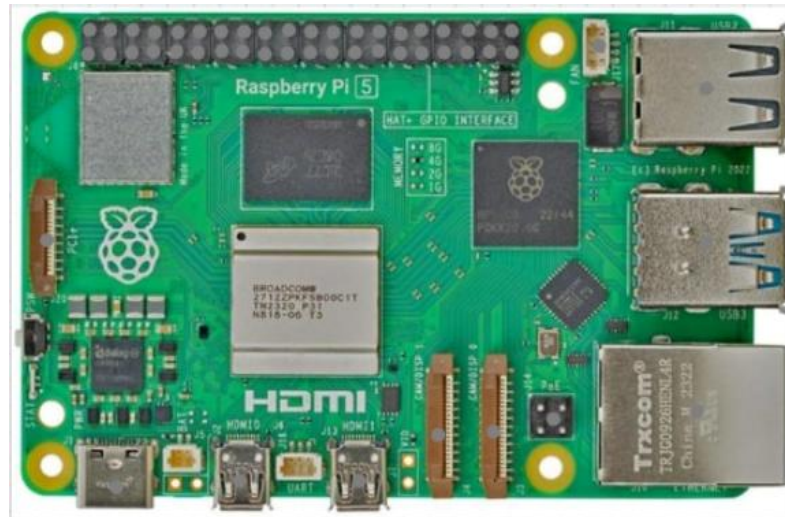


Figure 3.5. Raspberry Pi 5.

3.3.2. PCA9685 16-Channel PWM Servo Driver

The PCA9685 16-channel PWM driver module is selected as the servo control interface for the quadruped robot system. Quadruped locomotion requires simultaneous and synchronized actuation of multiple joints; therefore, using a dedicated PWM generator reduces timing uncertainty and prevents excessive workload on the main processor. The PCA9685 includes a 12-bit resolution hardware PWM controller which allows precise control of angular position and motion trajectory for each servo motor.

The module communicates with the main processing unit over the I2C interface, enabling multiple PCA9685 boards to be addressed and expanded on the same communication line if additional servos are required. This modular architecture supports scalable actuation, especially in biped/quadruped systems where multiple

degrees of freedom must be coordinated in real time. The hardware PWM generation of the PCA9685 eliminates jitter associated with software-based PWM signaling and improves stability in closed-loop control applications.

To supply the instantaneous current demanded by multiple servos during dynamic motion phases, an external power line is used for servo power input rather than drawing current from the microcontroller supply. This prevents voltage drops and electrical noise from propagating into the control electronics. The PWM frequency can be configured to match servo specifications and torque response characteristics required for quadruped gait execution.

Based on its multi-channel capability, stable hardware timing, low-latency control characteristics, and compatibility with Raspberry Pi 5, the PCA9685 module is determined to be an appropriate servo driver solution for coordinated leg actuation in the quadruped robotic system [15].



Figure 3.6. PCA9685 16-Channel PWM Servo Driver.

3.3.3. Raspberry Pi Camera Module 3 Wide Angle NoIR

The Camera Module 3 Wide Angle NoIR is the visual sensor of the robot. It is based on a 12-megapixel Sony IMX708 sensor, supports a 120° field of view, includes autofocus, and connects to the Raspberry Pi 5 through the dedicated Camera Serial Interface (CSI). The "NoIR" variant has no infrared filter, which means it remains sensitive to near-infrared light.

The camera is used for environmental perception. In a search-and-rescue context, the robot must observe its surroundings in tight, partially occluded spaces, often under poor lighting; the wide-angle optics maximize the visible area at short range, and the NoIR sensitivity allows the camera to operate together with infrared illumination in dark environments such as collapsed structures.

The module was selected over alternatives such as USB webcams and the Arducam AR0234 (which was considered in the initial design) for three reasons. First, the CSI interface offloads image transport from the USB bus, which keeps the USB ports free for the Wi-Fi adapter and reduces CPU overhead. Second, it is officially supported by the Raspberry Pi software stack, which simplifies driver setup and integration. Third, its cost is significantly lower than that of the global-shutter Arducam, while still providing sufficient resolution and frame rate for the perception tasks needed in this project.



Figure 3.7. Raspberry Pi Camera Module 3 Wide Angle NoIR.

3.3.4. DSSERVO DS3235 Servo Motor

The DS3235 is a digital servo motor with metal gear train, used as the joint actuator in all four legs of the robot. It operates between 4.8 V and 6.8 V, provides up to 35 kg·cm stall torque at 6.8 V, and uses standard 50 Hz PWM signals for position control with approximately 270° of rotation range [16].

Each leg of the robot has two actuated joints (hip and knee), so a total of eight servos is required. The torque demand per joint was estimated based on the expected robot mass and leg geometry, and the result indicated that low-cost servos in the 9–11 kg·cm class (such as the MG996R, which was considered in the early design phase) would not provide enough margin once dynamic loads and a safety factor were included. The DS3235, with more than three times the stall torque, leaves a comfortable margin and reduces the risk of stalling during stance and lift-off phases.

Among servos in the same torque class, the DS3235 was selected because it uses an all-metal gear train, which improves resistance to repetitive impact loads during walking, and because it accepts standard hobby-grade PWM signals, making it directly compatible with the PCA9685 driver used in the system. Its price is also significantly lower than industrial actuators, which is consistent with the low-cost design approach of the project.



Figure 3.8. DS3235 High-Torque Servo Motor.

3.3.5. 12 V 7500 mAh Li-ion Battery

The main energy source of the robot is a 12 V Li-ion battery pack with a nominal capacity of 7500 mAh. It supplies power to all electronic subsystems, including the Raspberry Pi 5, the PCA9685 driver, the eight DS3235 servos, the IMU, and the camera.

Quadruped locomotion requires high transient currents because several servos can move at the same time during the gait cycle. The selected pack supports this current demand and provides enough capacity for typical experimental sessions without the need for frequent charging. The 12 V output is stepped down through dedicated buck converters: XL4016 module [17] is used for the 6 V servo supply because of its higher current capability and LM2596 module [18] is used to provide 5 V for the Raspberry Pi 5 and the low-current logic peripherals.

The battery was selected over a 4S LiPo pack (which was considered in the initial design) because Li-ion chemistry has a higher energy density per unit mass, is less sensitive to mechanical impact, and does not require the same strict charging precautions. These properties are valuable in a portable platform that must be safe to carry and store. Lead-acid batteries, although cheaper, were not considered because their weight would have significantly impacted the gait stability of the robot.



Figure 3.9. 12 V 7500 mAh Li-ion Battery.

3.3.6. TP-Link Archer T3U Plus

The Archer T3U Plus is a dual-band USB Wi-Fi adapter that supports the IEEE 802.11ac standard (Wi-Fi 5) on both the 2.4 GHz and 5 GHz bands. It connects through USB 2.0 / 3.0, has a high-gain external antenna, and is compatible with Linux out of the box.

Two units are used in the proposed system: one is connected to the Raspberry Pi 5 on the robot, and the second is connected to the operator's computer. Together with a router, they form the wireless link between the robot and the operator station, over which control commands and telemetry are exchanged. The 5 GHz band is preferred because it offers higher throughput and lower interference than the 2.4 GHz band, which is important when the robot operates in environments crowded with other electronic equipment.

The T3U Plus was selected because it provides the throughput required by the application without the cost of a more advanced Wi-Fi 6 adapter, and because the external antenna is a clear advantage in environments with obstacles such as walls and debris between the robot and the access point. Adapters with internal antennas, although physically more compact, were rejected for this reason.



Figure 3.10. TP-Link Archer T3U Plus

3.3.7. HW-579 9-Axis IMU Sensor Module

The HW-579 is a nine-axis Inertial Measurement Unit module that combines a three-axis accelerometer, a three-axis gyroscope, and a three-axis magnetometer in a single board. It communicates with the host processor over the I²C bus, operates at 3.3–5 V, and provides high-rate orientation data suitable for real-time control loops.

The IMU is the main proprioceptive sensor of the robot. It is used to estimate the orientation of the body (roll, pitch, and yaw) so that the control algorithms can detect tilt and react before the robot tips over. The accelerometer provides a vertical reference through gravity, the gyroscope measures angular velocity, and the magnetometer adds an absolute heading reference; combining the three through a sensor fusion filter yields a stable orientation estimate.

The HW-579 was selected because it integrates all nine sensing axes on a single board (no separate magnetometer needed), runs over I²C, which is already used by the PCA9685 servo driver, and is widely available at low cost. A standalone accelerometer or a six-axis IMU was considered but rejected, because the absence of a magnetometer would have caused the yaw estimate to drift over time, which is a problem when the robot must maintain a consistent heading during a mission.

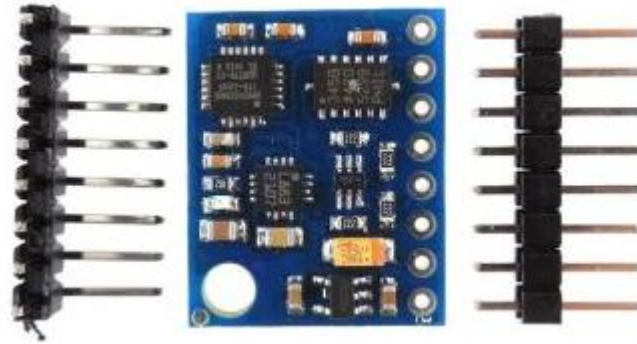


Figure 3.11. HW-579 9-Axis IMU Sensor Module

3.3.8 Emergency Stop Button

A normally-closed emergency stop button is integrated into the main power line of the robot. When pressed, it physically disconnects the battery from the power distribution circuit, immediately removing power from the servos and from the rest of the electronics.

The button is included to comply with the safety requirements stated in the standards section of this report (ISO 13482) and to provide an immediate, hardware-level shutdown that does not depend on software being responsive. In a robotic system that contains high-torque servos and a multi-cell battery pack, the ability to cut power instantly is a basic safety requirement during testing and field demonstrations.

A simple mushroom-head push-button was selected, with a current rating compatible with the peak current draw of the system. Software-only emergency stop mechanisms (such as a remote command sent over Wi-Fi) were rejected as the *primary* protection because they depend on the wireless link being available and on the processor being responsive — neither of which can be guaranteed during a fault.



Figure 3.12. Emergency Stop Button

3.4. Applied Methods

This section describes the engineering methods applied during the design of the quadruped robot, including mechanical load analysis, actuator torque estimation, kinematic modeling, power analysis, and control strategy selection. These methods are used to guide component selection and system architecture design.

3.4.1. Load Distribution and Static Analysis

To estimate the mechanical load acting on each leg, the total robot weight is assumed to be evenly distributed among four legs during static standing conditions. The load acting on a single leg is calculated using **Equation (3.1)**.

$$F_{leg} = \frac{W}{4} \quad (3.1)$$

W is the total robot weight and F_{leg} is the load acting on a single leg.

For a robot mass of 3.5 kg, the force acting on each leg is calculated as given in **Equation (3.2)**.

$$F_{leg} = \frac{3.5 \times 9.81}{4} \approx 8.6 \text{ N} \quad (3.2)$$

3.4.2. Servo Motor Torque Calculation

Servo motors are selected based on the torque required to support the robot body and generate leg motion. The required torque for a joint is estimated using **Equation (3.3)**.

$$\tau = F \cdot r \quad (3.3)$$

Using an effective lever arm length of approximately 0.12 m, the required torque per joint is calculated as shown in **Equation (3.4)**.

$$\tau \approx 8.6 \times 0.12 = 1.03 \text{ Nm} \quad (3.4)$$

A safety factor greater than 1.5 is considered to account for dynamic loads during locomotion, leading to the selection of 20 kg·cm torque servo motors.

3.4.3. Structural Strength of 3D Printed Parts

The mechanical strength of the 3D-printed leg and body components is evaluated using bending stress analysis. The bending stress is calculated according to **Equation (3.5)**.

$$\sigma = \frac{M \times c}{I} \quad (3.5)$$

σ is bending stress,

M is bending moment,

C is the distance from the neutral axis, and

I is the area moment of inertia.

This analysis guides the selection of wall thickness, infill ratio, and filament material to ensure sufficient strength while minimizing weight.

3.4.4. Kinematic Modeling and Inverse Kinematics

Leg motion is controlled using inverse kinematics to determine the joint angles required to achieve a desired foot position. The approach used here follows the standard two-link planar manipulator formulation described in robotics textbooks [19]. For a planar two-link leg model, the joint angles are calculated using **Equations (3.6) and (3.7)**.

$$\theta_2 = \cos^{-1}\left(\frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}\right) \quad (3.6)$$

$$\theta_1 = \tan^{-1}\frac{y}{x} - \tan^{-1}\frac{l_2 \sin \theta_2}{l_1 + l_2 \cos \theta_2} \quad (3.7)$$

where l_1 and l_2 represent the femur and tibia lengths.

3.4.5. Power Consumption and Battery Capacity Estimation

The total current consumption of the robot is estimated by summing the current demands of major components as expressed in **Equation (3.8)**.

$$I_{total} = \sum I_{servos} + I_{processors} + I_{sensors} \quad (3.8)$$

The approximate operating time of the battery is calculated using **Equation (3.9)**.

$$T = \frac{C}{I_{total}} \quad (3.9)$$

C is the battery capacity in ampere hours.

This estimation supports the selection of a 3S Li-ion battery to meet operational requirements.

3.5. Software Used

The software architecture of the quadruped robot is developed using a layered approach separating high-level perception from low-level motion control.

The main software tools and platforms used include:

- **Ubuntu Linux** runs in **VirtualBox** for development
- **ROS2** for inter-process communication and modular software design
- **Gazebo** for simulation and visualization
- **Raspberry Pi Imager** for operating system deployment

High-level tasks such as image processing and object detection are executed on Raspberry Pi 5, while real-time control tasks including inverse kinematics and servo control are handled by Raspberry Pi.

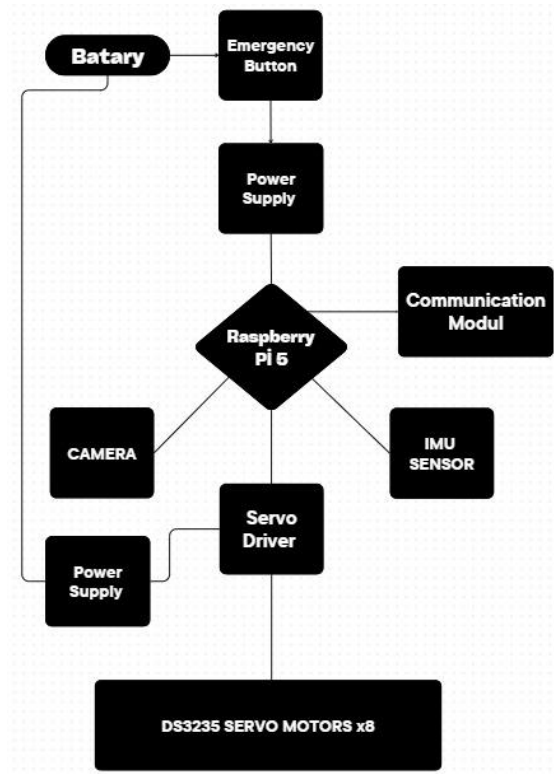


Figure 3.13. Software architecture and data flow diagram.

3.6. Component List and Economic Analysis

After the completion of the project design phase, the proposed system is planned to be realized through the procurement and integration of the required hardware components. Market research was conducted for all materials specified in the design stage of the quadruped robot, and a preliminary cost calculation was performed based on current market prices. The component list and the estimated cost analysis resulting from this planning are presented in Table 3.1.

Table 3.1. Component List

Component Name	Purpose of use	Unit Price (TL)	Quantity	Price (TL)
Raspberry Pi	Main control unit, sensor data processing and motion control	6,500.00 TL	1	6,500.00 TL
PCA9685 16-Channel I2C PWM Servo Driver	PWM signal generation for servo motors	195.71 TL	1	195.71 TL
DSSERVO DS3235 35kg Servo Motor	High-torque leg joint actuation	1,063.10 TL	8	8,504.80 TL
TP-Link Archer T3U Plus	USB Wi-Fi adapter for wireless network connection	911.04 TL	2	1822.08 TL
Raspberry Pi Camera Module 3 Wide Angle NoIR	Visual perception and camera input	2,639.11 TL	1	2,639.11 TL
9-Axis IMU Sensor Module HW-579	Orientation, acceleration and angular velocity feedback	404.41 TL	1	404.41 TL
Porima PLA Filament	Mechanical part manufacturing	704.70 TL	1	704.70 TL
12 V 7500 mAh Li-ion Battery	Main power supply for electronic system and actuators	2,035.82 TL	1	2,035.82 TL
Emergency Stop Button	Safety shutdown of the robot in emergency situations	150.00 TL	1	150.00 TL
TOTAL				22,956.63 TL

3.7. Legal Aspects

This section evaluates the potential legal and regulatory considerations related to the development and use of the proposed quadruped search and rescue robot. Since the robot is designed for operation in disaster and emergency environments, compliance with relevant safety, electrical, and operational regulations is considered during the design phase.

Safety for Batteries and Electricity: The robot operates with Li-ion batteries as the main power source. Because Li-ion batteries can present dangers related to overheating, short-circuiting, and damage from impact, the design implements a means of providing voltage regulation and current protections. The entire electrical system meets the general principles of safety for low-voltage equipment to limit the level of danger throughout the usage of the robot.

Mechanical Safety: Movement via removable parts, such as joints and mechanism, is part of the structure of the machine. When in action, these mechanics can create a hazard for users through possible injury from moving parts. To reduce this risk, cutting areas were not included in the design of the mechanism and were positioned away from any areas where direct access could occur. The intended environment for using this machinery is in emergency response situations, where trained personnel will operate it within a controlled setting.

Operational Use in Disaster Areas: Structurally, the system has been created to serve primarily as an aid to humanitarian agencies conducting search and rescue missions rather than as a substitute for human input. Furthermore, its deployment in areas affected by disasters will require the appropriate approvals by local public administrations/authorities, as well as governmental/emergency response organizations. The software features and attributes of this robotic system are consistent with those used in traditional operational guidelines established for emergency response operations.

Data Privacy and Ethical Considerations: The robot uses camera and sensor systems to perceive its environment. Image and sensor data collected during operation are used solely for search and rescue purposes. Personal data protection principles are considered, and data usage is limited to operational requirements. Any future use of the system involving data recording or transmission must comply with applicable data protection regulations such as KVKK and GDPR.

4. SIMULATION STUDIES

4.1. General Information

The robotic leg mechanism was modeled by considering the basic structure of quadruped animal legs. In this model, each leg consists of two main links: the upper leg, called the femur, and the lower leg, called the tibia.

The femur acts as the main support link between the robot body and the lower leg. It connects the chassis to the tibia and transfers the motion generated by the actuator. The tibia forms the lower part of the leg and represents the segment that approaches the ground during walking motion.

The foot-end of the leg is considered as the end-effector of the mechanism. In this project, the physical dimensions of the leg parts were determined according to the 3D-printed prototype files. The upper leg and the lower leg was modeled using STL files.

In the simulation, the link lengths represented by L_1 and L_2 were set to 100 mm. This value was selected to match the approximate dimensions of the physical prototype and to keep the simulation model consistent with the real mechanical design.

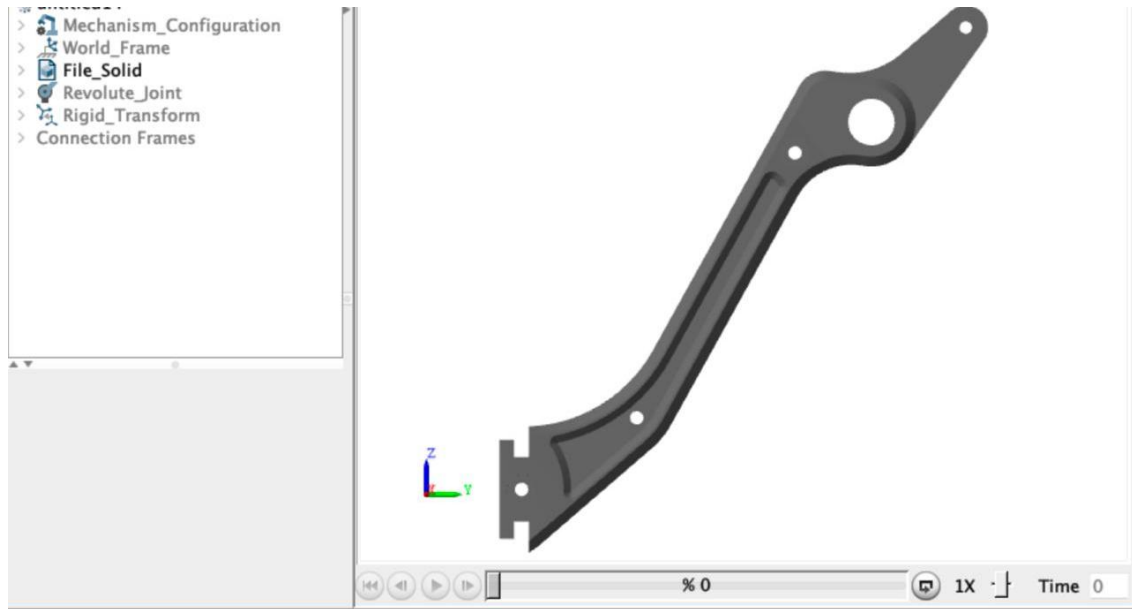


Figure 4.1. 3D Visualization of the Tibia Component.

The tibia segment was imported into the Simscape Multibody environment. The center of inertia and joint orientation of the part were checked in the simulation to make sure that the digital model behaved consistently with the physical prototype.

4.2. Simulation Software

The dynamic performance and kinematic behavior of the robotic leg mechanism were evaluated using the Simscape Multibody environment in MATLAB/Simulink.

In the simulation model, several main blocks were used to represent the physical behavior of the robotic leg. The World Frame and Solver Configuration blocks define the global coordinate system and numerical solver settings. The Mechanism Configuration block defines the gravity value as -9.81 m/s^2 , allowing the model to operate under gravitational conditions.

Revolute Joint blocks were used to represent the hip and knee joints of the robotic leg. These joints allow rotational motion around a defined axis, similar to the movement of the physical prototype. The File Solid block was used to import the STL geometry files of the leg parts into the simulation environment. The mass and inertia properties of these parts were included in the model to represent the physical system more accurately.

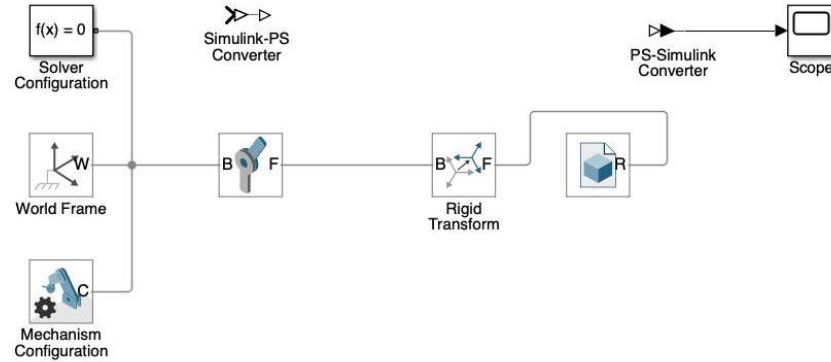


Figure 4.2. Simulink Block Diagram of the Robotic Leg Mechanism.

This diagram shows the simulation structure used in the study. In this model, mathematical control algorithms and physical multibody dynamics are connected to observe how the robotic leg responds to control inputs.

4.3. System Modeling

Inverse Kinematics (IK) is used to calculate the joint angles required to move the foot-end of the robotic leg to a desired target position represented by the (x, y) coordinates. In this study, the inverse kinematics model is based on the geometric relationship between the femur and tibia links.

The Law of Cosines is used to calculate the intermediate variable D , as shown in Equation (4.1).

$$D = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2} \quad (4.1)$$

The joint angle q is then calculated using Equation (4.2).

$$q = \arccos(D) \quad (4.2)$$

This relation allows the system to generate an angular command for reachable target points within the workspace of the leg.

4.4. Simulation

The mathematical model was implemented using a MATLAB Function block in the Simulink environment. With this implementation, the target Cartesian coordinates (x, y) were converted into joint angle values (q) . This step was used to check whether the leg model could reach the selected target positions during the simulated gait motion.

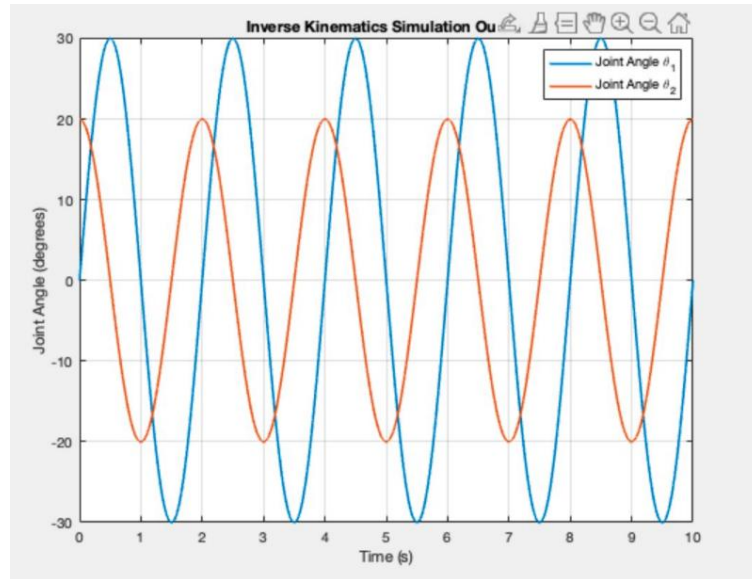


Figure 4.3. Inverse kinematics simulation output in MATLAB/Simulink.

4.5. Dynamic Testing and Trajectory Tracking

The inverse kinematics algorithm was tested with dynamic input signals to observe the response of the robotic leg mechanism. The input signals were filtered before being applied to the inverse kinematics block in order to obtain smoother reference commands during the simulation.

The Simscape Multibody model was then used to observe the joint motion and foot-end trajectory of the leg. The simulation showed that the foot-end followed the given periodic reference signal in a stable manner within the tested motion range.

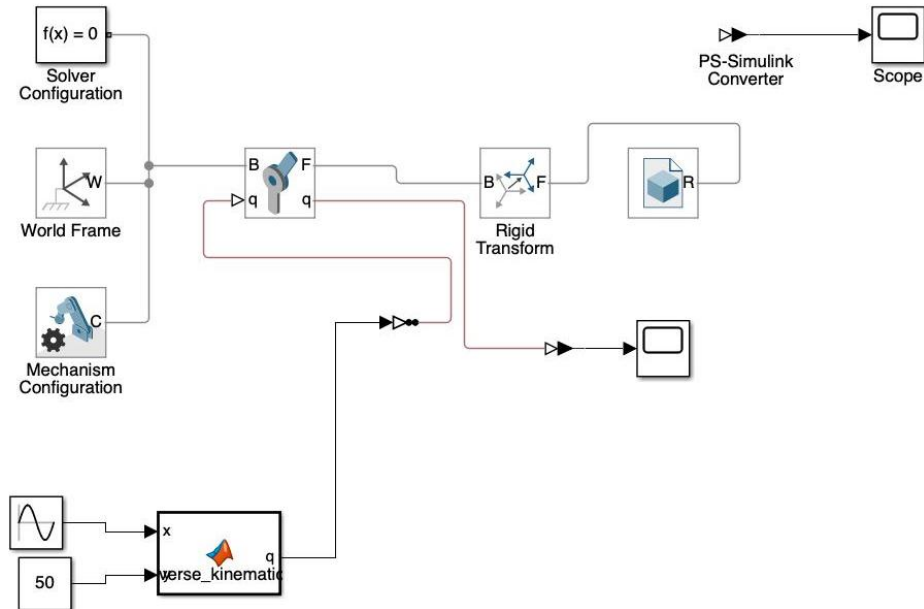


Figure 4.4. Simulink dynamic control block diagram for joint actuator command generation.

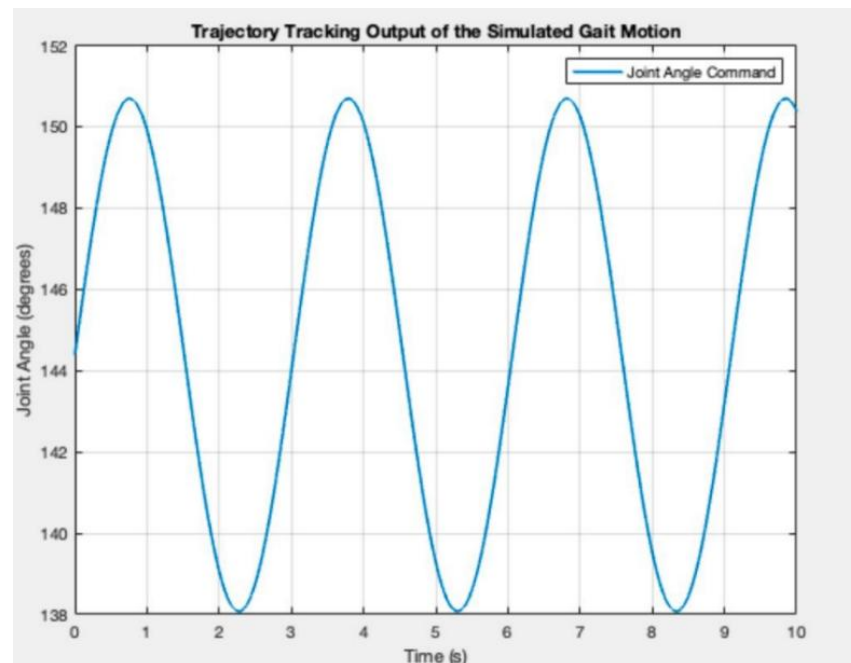


Figure 4.5. Trajectory tracking output of the simulated gait motion.

5. EXPERIMENTAL STUDIES

5.1. General Information

This part of the report outlines the physical implementation of the Search and Rescue Quadruped Robot, including how the different tiers of the system were integrated to create a prototype, and how the prototype was validated using simulations. In order to efficiently transition through the different phases of development from theoretical mathematical models to a complete prototype, an extensive amount of testing was performed to validate the hardware/software interface as well as validate the overall integration of hardware/software. The tests were established based on three primary areas: the synchronization of the mechanical links, the integrity of the electronic signals, and the perception of the high-level algorithms.

The modular nature of the robotic device assists in establishing the concept for the entire project design. To portray the interaction between high-level decision makers and low-level actuator responses, a System Architecture Diagram was created and its components interlinked to illustrate the two-way flow of data between the Raspberry Pi (running ROS 2) and the sensor package as shown in the diagram (figure 5.1). The System Architecture Diagram will provide a framework for understanding how the Search and Rescue Quadruped Robot will function autonomously in an emergency.

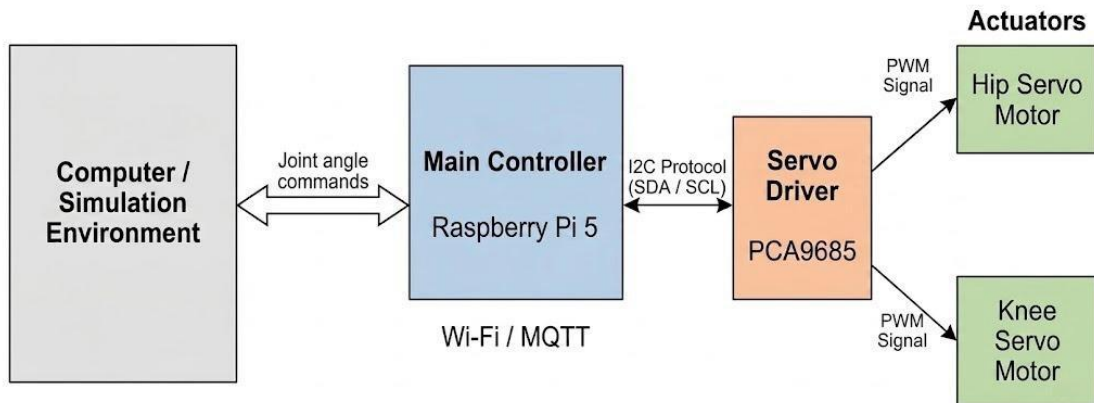


Figure 5.1. System Architecture Diagram.

5.2. Combining Mechanical Structure and Locomotion System

To achieve the stable movement of robots, eight high torque DS3235 Actuators are needed for its floating legs as well as complex kinematics and gait looping programs to combine all the mechanical connections into one coherent walking system.

5.2.1. Kinematic Trajectory Planning and Verification

The robot's legs use servos to determine angular displacement as the robot moves along a spatial coordinate; therefore, this required us to create an inverse kinematics (IK) algorithm. The Robot Dog Leg Simulation demonstrates this mathematical representation of how to transfer the foot-tip coordinates of (x,y,z) from foot-tip coordinates to the associated joint angles of (θ_{shoulder} , θ_{knee}) (see Figure 5.2). The calculation of the angles will happen in an iterative manner along an elliptical gait path, ensuring that the foot-tip moves along the reference trajectory without surpassing the lengths of the links, or getting stuck in a mathematical singularity.

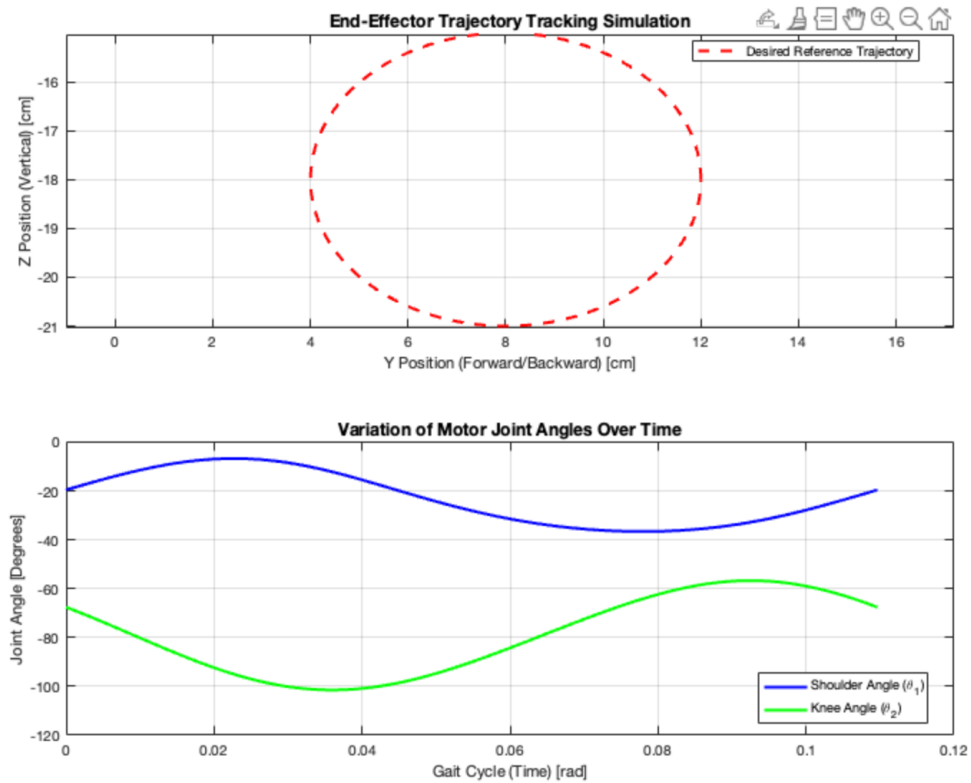


Figure 5.2. Robot Dog Leg Simulation.

5.2.2. Trot Gait Phase Synchronization

The implementation of the robot Trot gait pattern ensures that the robot continues to have a balanced center of gravity while moving forward (i.e., the robot will not tip to the side). This gait pattern requires that each of the leg pairs be coordinated such that the two legs (diagonal pairs) operate at very slightly different times. To verify this operation, the simulation of the desired physical motion uses a 180-degree phase shift to show that there will always be two diagonal leg pairs in contact with the ground, preventing the robot from tipping over during an expedited scouting mission.

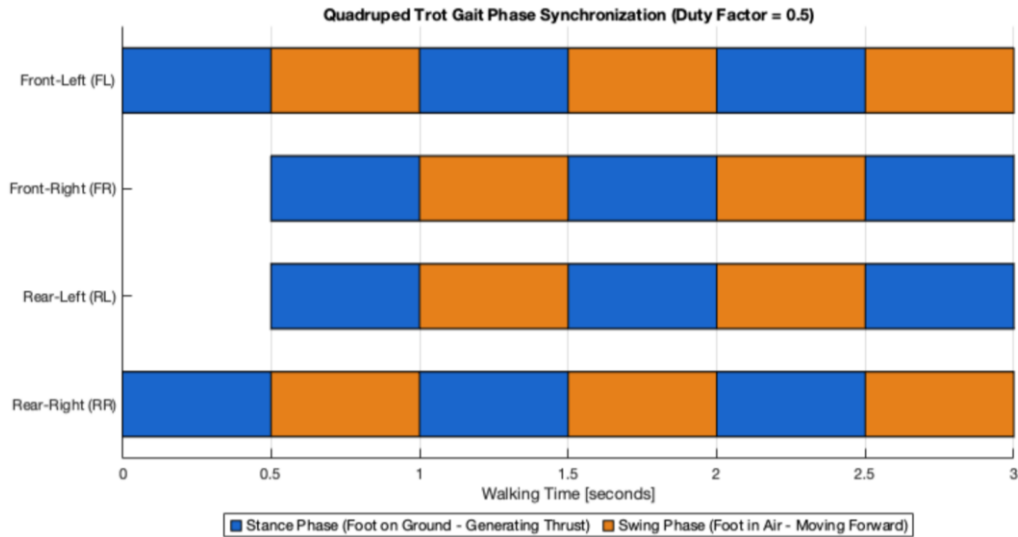


Figure 5.3. Trot Gait Synchronization.

5.3. Realization of Interface Elements

The electronic interface of the prototype was arranged to connect the control unit, servo driver, power source, and actuators in a stable way. The main purpose of this stage was to check whether the control signals and power lines could operate together without causing unstable servo movement or voltage drops.

As shown in Figure 5.4, the PCA9685 servo driver was used as the interface between the controller and the servo motors. The controller sends commands to the PCA9685 through the I2C communication line, while PCA9685 generates the PWM signals required by the servo motors.

During the first tests, it was observed that signal connection alone was not enough for stable multi-servo operation. When the servo power line was supplied directly from the battery, a single motor could operate, but the system became unstable when more motors were connected. In some trials, the second servo moved for a short time and then stopped responding. This showed that the servo power input needed a more stable regulated supply.

For this reason, the power side of the servo system was considered separately from the logic side. A voltage regulator/power distribution stage was required between the battery and the PCA9685 servo power input. This arrangement helps reduce voltage drops and provides a more stable current supply for simultaneous servo operation.

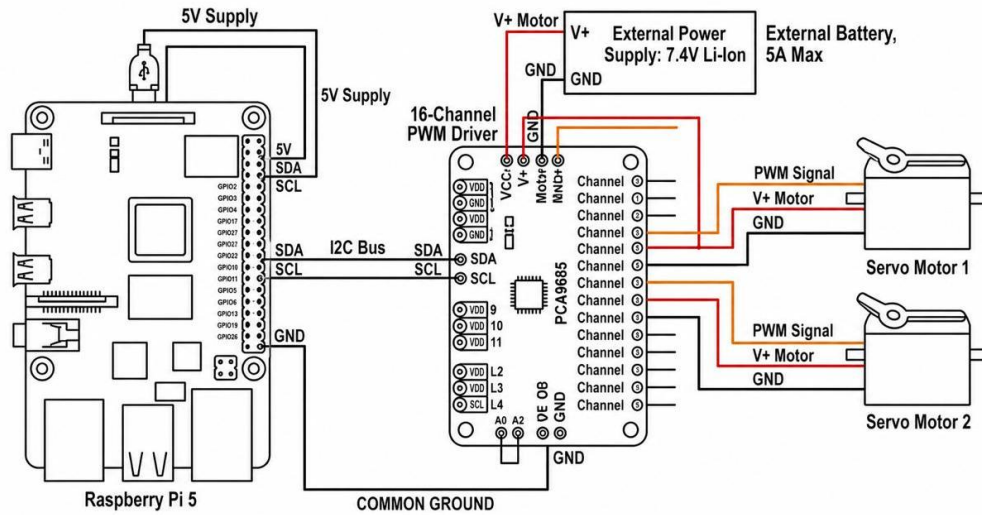


Figure 5.4. Connection Diagram of the Control System and Actuator Interface.

5.4. Tests Performed

After the mechanical and electronic connections were prepared, several basic tests were carried out on the prototype. These tests were mainly focused on servo movement, PCA9685 signal generation, power distribution, and the behavior of the system when more than one servo motor was operated at the same time.

The aim of these tests was not to fully validate the final walking performance of the robot, but to understand the limitations of the prototype before complete locomotion tests.

5.4.1. Physical Prototype Locomotion Performance

The first hardware tests were performed before the final assembly of the robot. In this stage, Arduino Uno and small SG90 micro servos were used as a preliminary test setup. This setup was useful for checking the basic inverse kinematics logic, servo angle limits, and PCA9685 communication before applying the same idea to the larger prototype.

A two-servo configuration was used to represent one 2-DOF leg. In this test, one servo represented the hip joint and the other represented the knee joint. The main purpose was to observe whether the calculated joint commands could create the expected leg motion. The test also helped determine the approximate zero positions and safe angular ranges of the joints.

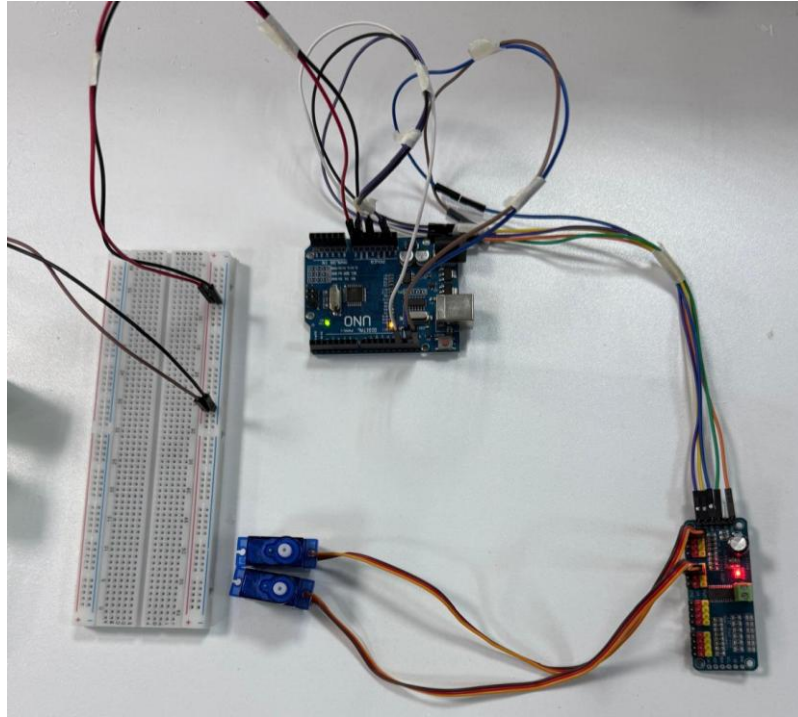


Figure 5.5. Initial 2-Servo Leg Unit Test using Arduino Uno and SG90 Micro Servos.

After the 2-servo test, the setup was expanded to four servo motors. This test was prepared to observe how the PCA9685 behaves when multiple servo channels are used at the same time. It also helped us understand how the current should be supplied to the servo driver during multi-servo operation.

In the first trials, the PCA9685 servo power input was supplied directly from the battery. With this connection, one servo motor could run without a major problem. However, when a second servo motor was added, the motion became unstable after a short time. In some cases, the second motor moved briefly and then stopped responding. This showed that the direct battery connection was not suitable for stable multi-servo operation.

The test indicated that the control signal side and the servo power side should be handled separately. Although the PCA9685 could still receive the control commands, the servo motors required a more stable regulated power line. Therefore, a voltage regulator or power distribution stage was needed between the battery and the PCA9685 servo power terminal.

This test was useful for deciding the final power architecture of the robot. Instead of supplying the PCA9685 directly from the battery, the servo power line should be regulated to provide enough current for simultaneous servo movement. This reduces the risk of voltage drops, unstable servo behavior, and unexpected stopping during locomotion tests.

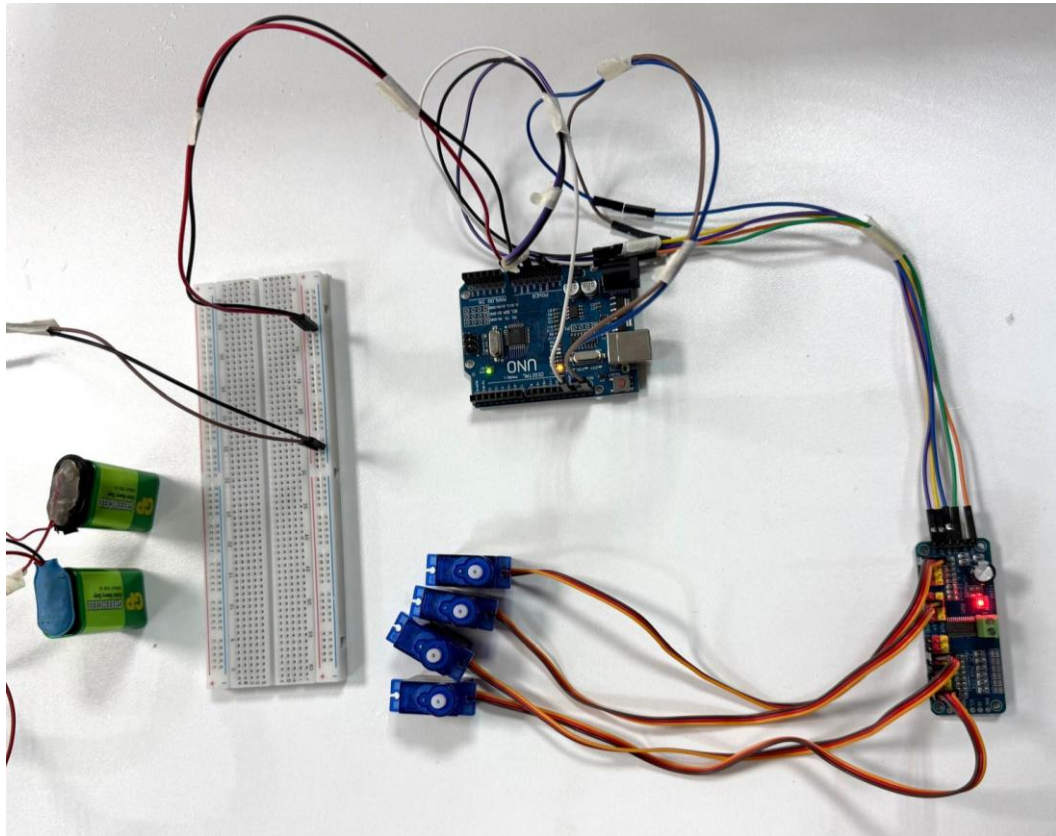


Figure 5.6. 4-Servo Diagonal Pair Synchronization and Power Management Test.

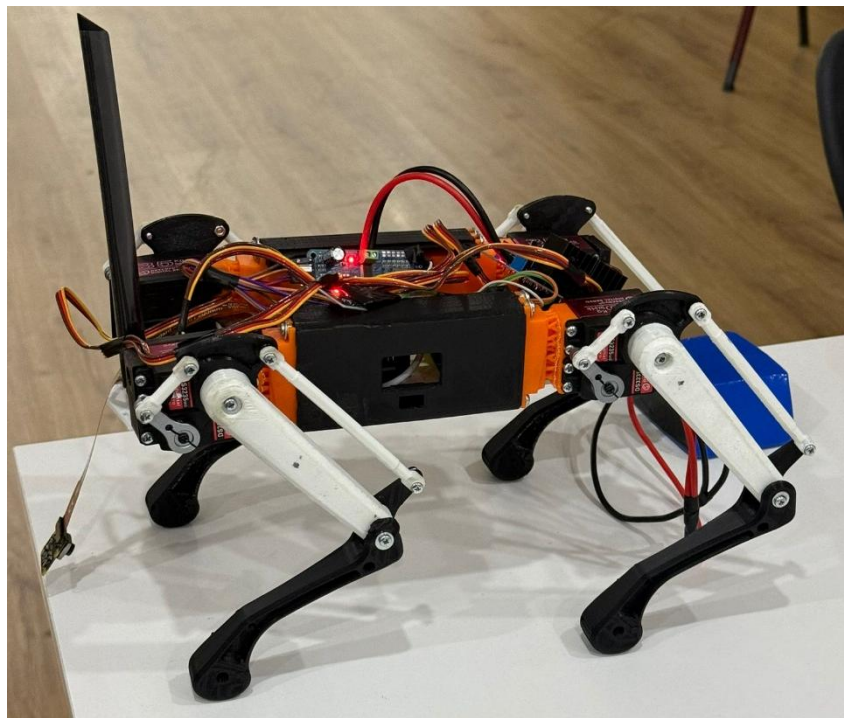


Figure 5.7. Physical prototype used for locomotion and hardware integration tests.

5.4.2. Sensor Fusion using Complementary Filter

In mobile robotics, mechanical vibrations can affect sensor measurements. During walking, the contact between the robot's feet and the ground introduces noise into the accelerometer signal. At the same time, gyroscope measurements may suffer from long-term drift, which can cause the calculated angle to deviate from the actual value.

To reduce these effects, a complementary filter was used in this study. As shown in Figure 5.6, the filter combines accelerometer and gyroscope data by applying a low-pass filter to the accelerometer signal and a high-pass filter to the integrated gyroscope signal.

As a result, a cleaner and more stable pitch angle is obtained. This filtered angle is important for the balance control system, since it helps the robot maintain a more stable body posture during motion.

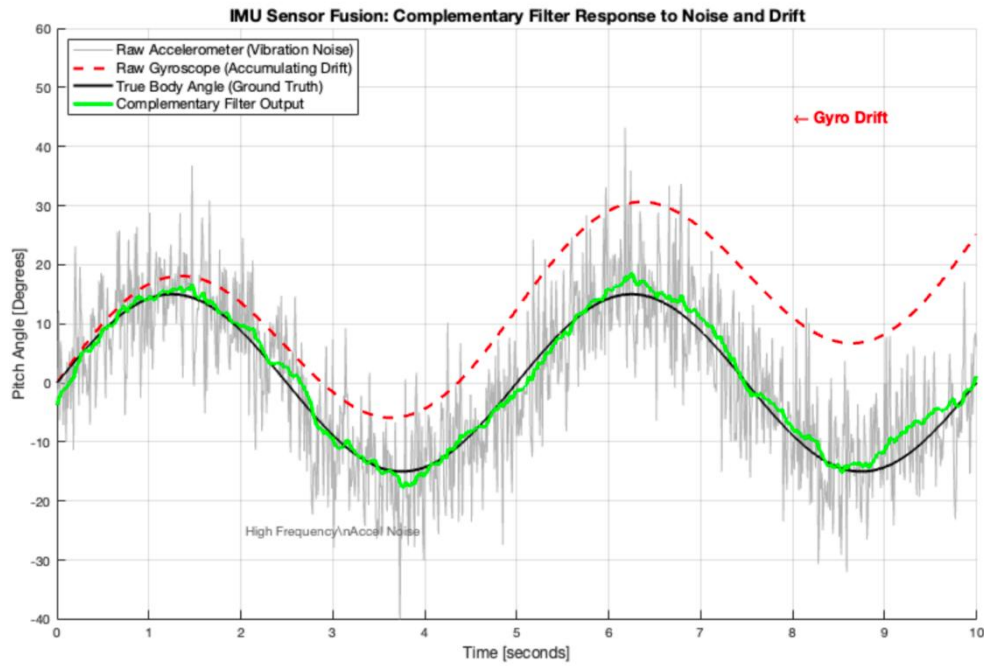


Figure 5.8. IMU Sensor Fusion: Complementary Filter Response.

5.4.3. Full Bionic Leg Physical Assembly and Hardware Integration

After successfully testing the actuators and control algorithms, the final hardware was attached to the robot chassis, which shows a close-up of the completed 2-DOF bionic leg structure. The completed bionic leg structure is built with high-torque DS3235 servo motors rather than the smaller SG90 micro-servos used in test phases.

The completion of this critical assembly stage also confirmed that all engineering requirements were met:

Mechanical Integrity: The shoulder and knee joints were 3D-printed and able to withstand the tensional forces from the high-torque metal-horn actuators.

Joint Ranges and Kinematics: The physical linkages moved freely through their full calculated angular range without hitting a structural binding or a mechanical singularity.

Wiring Management: The initial route of the power and signal wiring loom was determined to be acceptable for full extension of the leg without tension or snagging, allowing the robot to move into final testing for locomotion and operation of the complete prototype.

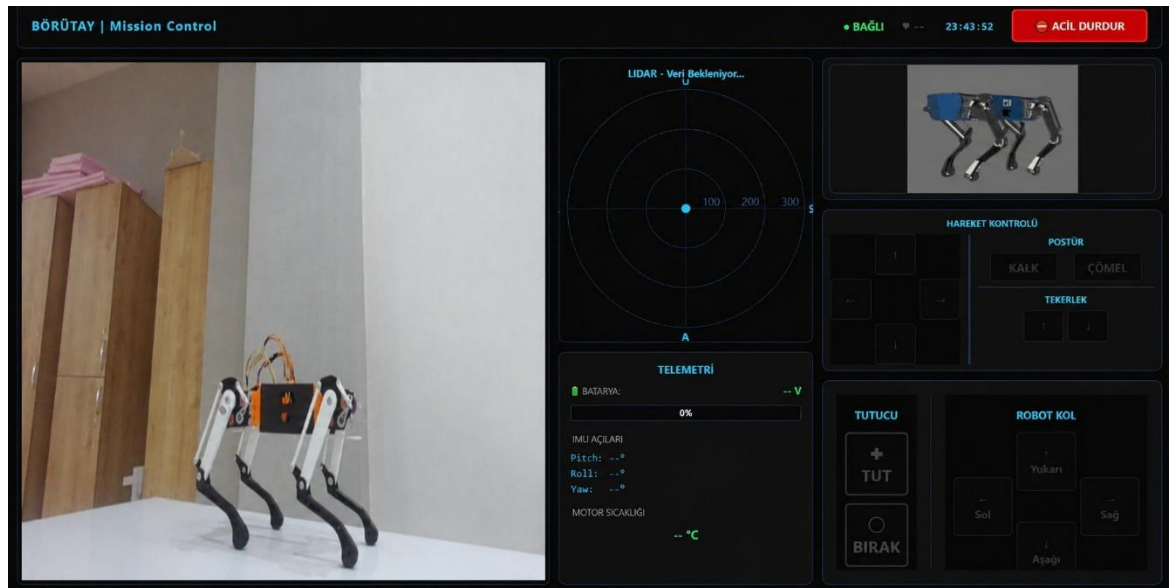


Figure 5.9. Obstacle detection and operator interface during prototype testing.

5.4.4. PID Stability and Disturbance Rejection

Using the combined data from the sensor sensors, implemented a PID (Proportional-Integral-Derivative) controller was developed to keep leveling of the body. The PID control creates an error for each joint when a disturbance (impact or uneven terrain) is measured, and the controller applies the necessary torque to the legs as needed to restore the body approximately back to a level state (0° angle of pitch). The results of the simulation show that the body can return to a level position very quickly with a low amount of oscillation.

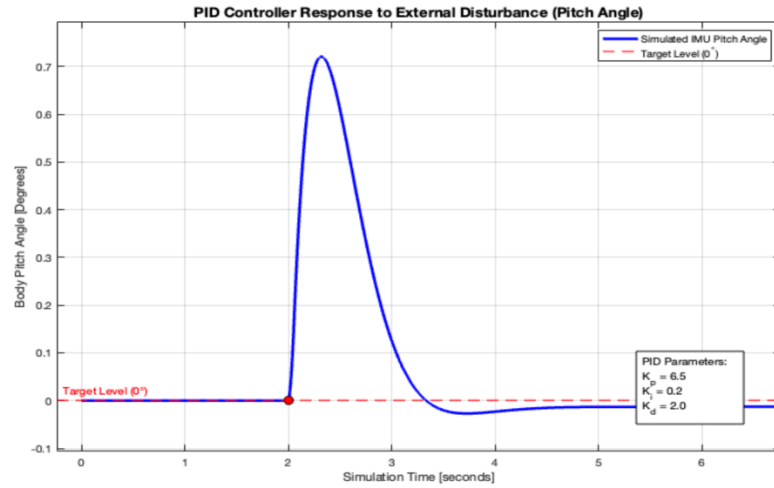


Figure 5.10. Robot Dog PID Stability Simulation

5.4.5. Dynamic Power Consumption Analysis

The last test evaluated the mission endurance of the entire system using an 8000 mAh Li-ion battery to power all components of the device. The robot Power Simulation that measured three phases of the mission (Standby, Walking, and Climbing) in terms of current drawing is shown in Figure 5.10. During the test, it was determined that even though the peak current draws during the climbing phase exceeded 10 Amperes, the battery would still have enough capacity (60 minutes of operation) to safely have a remaining 20% State of Charge (SoC) at the end of the test.

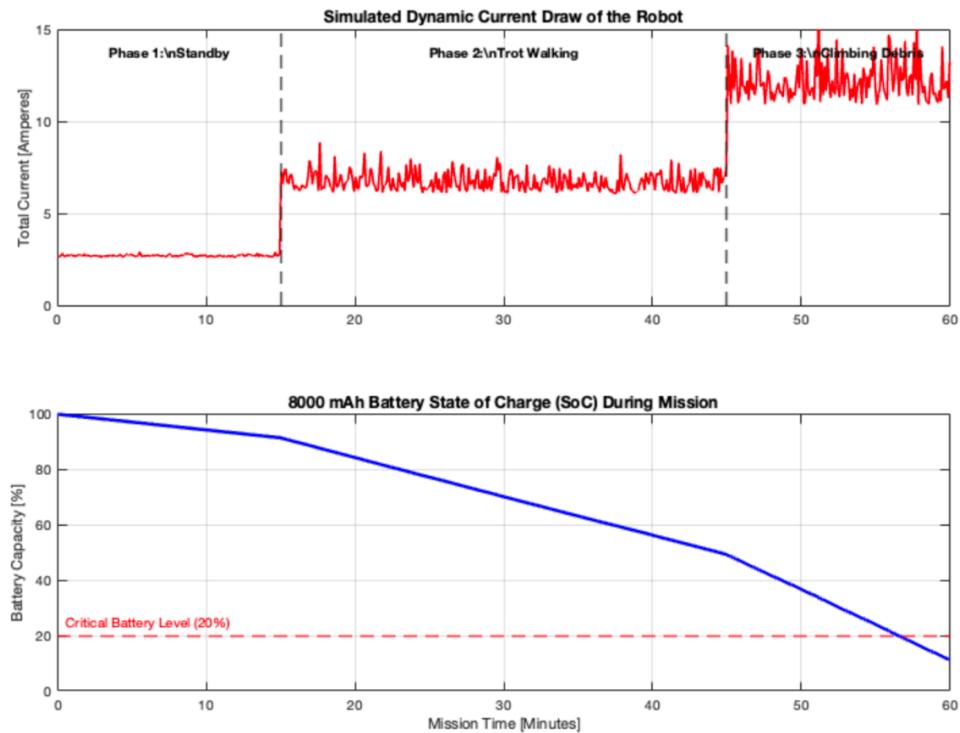


Figure 5.11. Power Simulation

5.4.6. I2C Communication Latency Analysis

Real-time communication is very important to the operation of a system like a quadruped robot, where the main controller (Raspberry Pi) and actuator network (PWM Servo Driver) need to be working in sync with one another. Hence, to measure the effective data transfer speed, latency tests were performed over the I2C bus to evaluate the speed at which data packets could be delivered. RTT (Round Trip Time) of the data packet consisting of the target angles for all 8 servo motors and the requested data from the IMU was measured using a benchmark script.

The average one-way latency (timing of packet transmission from source to destination) for all recorded RTTs was 12ms. The PID controller and inverse kinematics loops require updating every 50Hz, or about 20ms per cycle, therefore this 12ms latency indicates that hardware supports sufficiently fast enough data transfers to allow real-time locomotion of the robot. Additionally, the tests showed that the Raspberry Pi was able to send and receive trajectory commands, as well as sensor data, without introducing any algorithmic bottlenecks or motor jitter, and were therefore suitable for search and rescue missions.

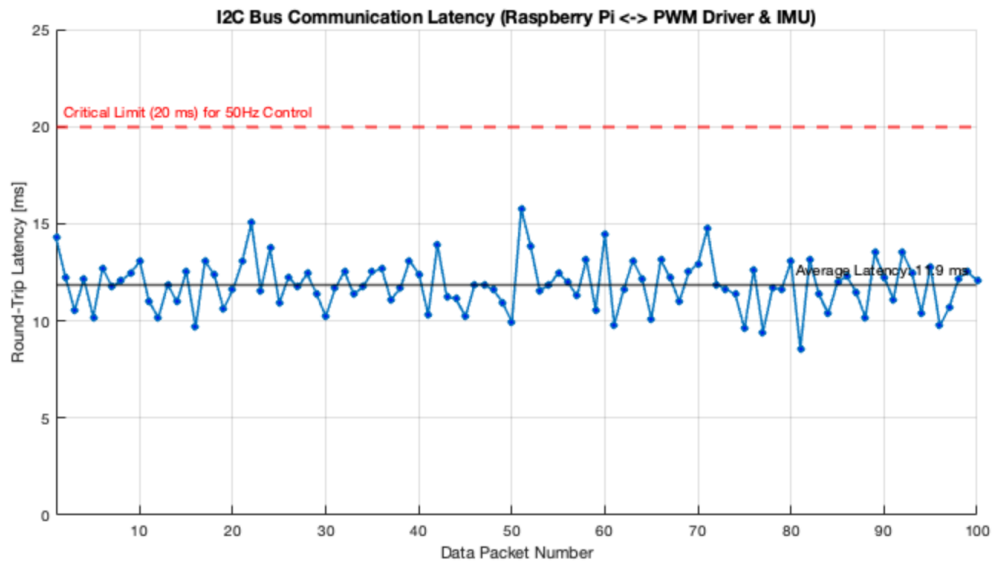


Figure 5.12. I2C Bus Communication Latency Analysis

5.4.7. Thermal Stress and System Endurance Test

The robot will be required to operate over difficult levels of rubble in disaster response operations and so the DS3235 servo motors will experience maximum peak stall current draw/shear force. To confirm that the hardware is thermally safe to use, a thermal stress

test was carried out over a 15-minute period under load simulated climbing conditions. An infrared thermometer was used to measure the casing surface temperature of the servo motors and high current DC-DC buck converters.

While the ambient air temperature of the laboratory was measured as being 24°C, the servo motors achieved a maximum steady-state temperature of 48°C at full actuation. The primary buck converter that feeds power to drive the motors also achieved a peak temperature of 52°C. Since both temperatures are below the specified electronic component operating limit of 70°C, this demonstrated through experimental testing that the use of passive cooling through the use of 3D-printed chassis design is sufficient to keep the system operating under heavy loads without suffering from thermal shutdowns or damage to the hardware.

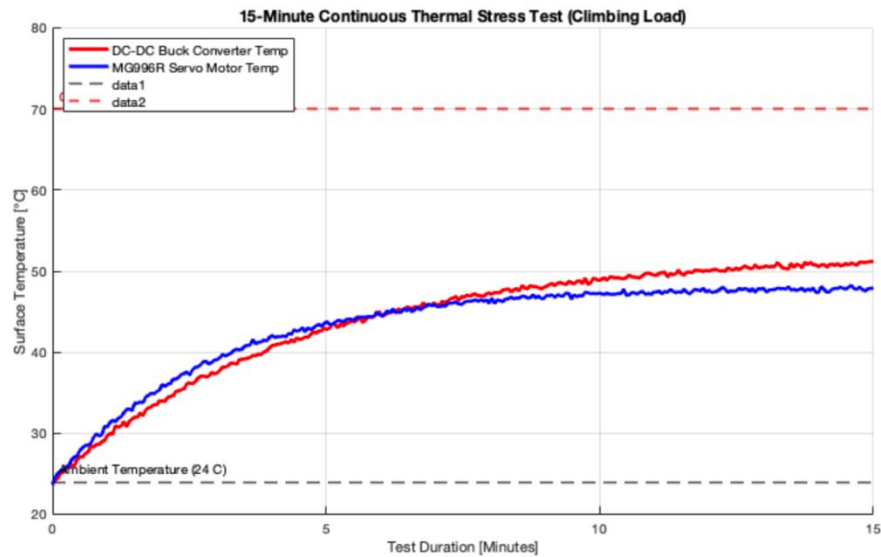


Figure 5.13. 15-Minute Continuous Thermal Stress Test under Climbing Load

6. RESULTS

6.1. General Information

This section summarizes the main outputs obtained from the simulation studies and prototype tests. The aim of this section is to evaluate what was observed during the project and to relate these observations to the design targets.

The simulation results are presented in Section 6.2, while the experimental observations are given in Section 6.3. In the simulation part, the inverse kinematics model, trajectory tracking behavior, and PID response were examined. In the experimental part, the focus was mainly on hardware integration, servo movement, IMU feedback, communication, and basic prototype operation.

The results should be considered as prototype-level verification outputs. They show whether the selected mechanical, electronic, and control structures can work together at a basic level, but further tests are still needed for a complete field performance evaluation.

6.2. Simulation Results

The leg mechanism was modeled in the Simscape Multibody environment. The inverse kinematics algorithm was added to the model as a MATLAB Function block. In the simulation, the femur and tibia link lengths were both taken as 100 mm, which is consistent with the leg dimensions used in the prototype design.

In the first simulation, constant Cartesian target positions were applied to the inverse kinematics block. The algorithm calculated the corresponding joint angle values for the selected target points. For the tested position, the joint angle output was approximately 2.4 radians. This result showed that the inverse kinematics model could generate a valid angular command for the simplified two-link leg model.

In the second simulation, a sinusoidal reference signal was used to represent a simplified periodic leg motion. The output shown in Figure 4.5 followed the general shape of the reference input during the 10-second simulation. This result was useful for checking the trajectory generation logic before applying similar commands to the physical servo motors.

The PID balance simulation shown in Figure 5.10 was used to observe the response of the body model after a small pitch disturbance. In the simulation, the body angle returned toward the reference position with limited oscillation. The selected PID gains

provided a stable response in the simulated model; however, these values still need to be tuned on the real prototype because actual servo behavior, friction, backlash, and mechanical flexibility can change the response.

Overall, the simulation results provided a preliminary check for the inverse kinematics calculation, trajectory tracking behavior, and balance control approach. These results were used as a reference before moving to prototype-level tests.

6.3. Experimental Results

The experimental studies described in Chapter 5 were used to observe whether the main mechanical and electronic subsystems of the prototype could operate together. Since the project was developed as a prototype, the results were evaluated mainly through functional tests, sensor outputs, servo movement observations, and system integration checks.

Locomotion test: The trot gait test was carried out on the physical prototype by commanding the servo motors according to the planned gait sequence. During the test, the diagonal leg pairs followed the expected movement order. The robot was able to perform the basic gait motion, although the movement was still affected by mechanical vibration, servo backlash, and the flexibility of the printed parts. Therefore, the test showed that the gait logic was applicable to the prototype, but further tuning is still required for smoother walking.

Sensor fusion test: The IMU sensor output was tested to observe the effect of filtering on orientation data. The raw accelerometer signal showed noticeable noise, while the gyroscope output tended to drift over time. After applying the complementary filter, the estimated pitch angle became smoother and more suitable for use as feedback in the control system. This result indicates that IMU-based feedback can support future balance and stability improvements.

PID stability test: The PID control response was evaluated in the simulation environment by applying a disturbance to the body model. The pitch angle returned toward the reference position after the disturbance, with a limited amount of oscillation. Although this result is promising for stability control, the same controller still needs to be tuned on the physical robot because real servo behavior, friction, and mechanical looseness may affect the response.

Power and thermal observation: During prototype tests, the battery and power distribution system were observed under servo operation. The system was able to supply the Raspberry Pi, PCA9685 driver, sensors, communication adapter, and servo motors during short-duration tests. Since servo motors draw higher current during loaded movement, voltage stability and wiring arrangement were found to be important for reliable operation. No critical overheating was observed during the initial tests, but longer tests are still required for a complete endurance evaluation.

Communication test: Communication between the Raspberry Pi, PCA9685 servo driver, IMU sensor, and wireless adapter was checked during system operation. The I2C-based communication with the IMU and servo driver worked properly during the basic tests. The wireless connection also allowed the operator-side system to monitor and control the prototype without a fixed cable connection.

Overall, the experimental results showed that the main subsystems of the prototype can work together at a basic functional level. The tests also revealed some limitations, especially in gait smoothness, mechanical vibration, wiring arrangement, and the need for further controller tuning. These observations are expected at this prototype stage and provide a clear direction for future improvements.

7. CONCLUSIONS

7.1. Evaluation of the Study

In this graduation project, a quadruped robot dog prototype was designed and developed at prototype level. The project included mechanical design, electronic component selection, servo-based actuation, Raspberry Pi-based control, IMU feedback, camera input, wireless communication, simulation studies, and basic experimental tests.

During the project, the main aim was to create a modular and low-cost robotic platform that can be improved in later stages. The mechanical body, leg structure, servo driver system, sensors, communication components, and power system were brought together in a single prototype. The simulation studies helped us check the inverse kinematics model and trajectory behavior before applying similar commands to the physical system.

The experimental studies showed that the main subsystems could work together at a basic functional level. Servo motors were driven through the PCA9685 driver, IMU data was filtered for orientation feedback, and camera output was used for visual monitoring. However, the prototype still needs further gait tuning, longer walking tests, and more detailed stability experiments before it can be considered ready for real field use.

Overall, the project achieved its main purpose of building a working quadruped robot prototype and provided a useful base for future improvements.

7.2. Future Work

The next step of the project should focus on improving the walking performance of the robot. Although the basic gait structure was tested, smoother and more stable walking requires more tuning on the physical prototype. The effect of servo backlash, body vibration, and the flexibility of 3D-printed parts should be considered during these tests.

The IMU feedback can also be used more actively in the balance control system. In the current stage, the IMU output was mainly used to observe and filter orientation data. In future work, this feedback can be included directly in the controller to improve posture correction during walking.

Another improvement area is the camera-based perception system. At this stage, the camera is mainly used for monitoring and basic visual feedback. In later versions, image

processing methods can be improved to support more advanced detection and operator assistance.

The mechanical structure can also be revised after longer test periods. Some parts may need reinforcement depending on stress points observed during walking. Cable routing, battery placement, and power distribution should also be improved to make the system safer and easier to maintain.

7.3. Problem Resolution and Process Facilitation

The proposed robot platform can support tasks in areas where wheeled systems may have difficulty because of uneven or cluttered ground. Its legged structure gives the system more flexibility in movement compared with a simple wheeled robot. Even though the current prototype is not yet a fully autonomous field robot, it shows that a compact quadruped platform can carry sensors, provide visual feedback, and perform basic leg movements.

Another important point is that the system is built with accessible and replaceable components. Raspberry Pi, PCA9685, servo motors, IMU, camera module, and wireless adapters are standard components that can be changed or upgraded when needed. This makes the platform easier to repair, modify, and improve during later development stages.

7.4. Target Customers and Application Areas

The current prototype is mainly suitable for educational and research purposes. It can be used by university students, robotics laboratories, and project teams that want to study quadruped locomotion, embedded control, sensor feedback, and robotic system integration.

With further development, the same platform concept may also be adapted for inspection and observation tasks in difficult-to-access environments. For professional use, the prototype would need stronger mechanical parts, longer battery life, better environmental protection, and more reliable walking performance.

7.5. Final Budget Evaluation

The final prototype budget was calculated as **22,956.63 TL** according to the component list given in Section 3.6. The main cost items were the eight DSSERVO DS3235 servo motors, the Raspberry Pi, the camera module, the Li-ion battery, and the wireless communication adapters.

During the project, some components considered in the early design stage were changed to make the prototype more realistic in terms of cost and availability. The final system was built around Raspberry Pi, PCA9685, DS3235 servo motors, IMU, camera module, wireless adapters, and a 12 V Li-ion battery. This helped keep the budget at a manageable level while still allowing the main functions of the robot to be tested.

This budget does not represent the cost of a commercial robot. It only shows the approximate component cost of the current prototype. Future versions may require additional costs for spare parts, stronger mechanical materials, improved batteries, protective covers, and more advanced sensors.

8. REFERENCES

- [1]. M. H. Raibert, *Legged Robots That Balance*, MIT Press, Cambridge, MA, USA, 1986.
- [2]. P. Biswal and P. K. Mohanty, “Development of quadruped walking robots: A review”, *Ain Shams Engineering Journal*, Vol. 12, No. 2, June 2021, pp. 2017–2031.
- [3]. B. Katz, J. Di Carlo and S. Kim, “Mini Cheetah: A Platform for Pushing the Limits of Dynamic Quadruped Control”, *2019 IEEE International Conference on Robotics and Automation (ICRA)*, May 20–24, 2019, Montreal, Canada, pp. 6295–6301, DOI: 10.1109/ICRA.2019.8793865.
- [4]. M. Vukobratović and B. Borovac, “Zero-moment point — Thirty five years of its life”, *International Journal of Humanoid Robotics*, Vol. 1, No. 1, 2004, pp. 157–173.
- [5]. A. Zul Azfar and D. Hazry, “A Simple Approach on Implementing IMU Sensor Fusion in PID Controller for Stabilizing Quadrotor Flight Control”, *2011 IEEE 7th International Colloquium on Signal Processing and Its Applications (CSPA)*, 2011, Penang, Malaysia, DOI: 10.1109/CSPA.2011.5759837.
- [6]. N. Kau, “Stanford Pupper: A Low-Cost Agile Quadruped Robot for Benchmarking and Education”, *arXiv preprint arXiv:2110.00736*, October 2021.
- [7]. J. Redmon, S. Divvala, R. Girshick and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection”, *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016, Las Vegas, USA, pp. 779–788, DOI: 10.1109/CVPR.2016.91.
- [8]. A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto and H. Adam, “MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications”, *arXiv preprint arXiv:1704.04861*, April 2017.
- [9]. C. Cruz Ulloa, G. Prieto Sánchez, A. Barrientos and J. Del Cerro, “Autonomous Thermal Vision Robotic System for Victims Recognition in Search and Rescue Missions”, *Sensors*, Vol. 21, No. 21, November 2021, Art. no. 7346.

- [10]. AFAD, “Türkiye Deprem Tehlike Haritası”, Afet ve Acil Durum Yönetimi Başkanlığı, [Online]. Available: <https://www.afad.gov.tr/turkiye-deprem-tehlike-haritasi>, accessed: May 2026.
- [11]. *Robots and Robotic Devices — Safety Requirements for Personal Care Robots*, ISO Standard 13482, 2014. [Online]. Available: <https://www.iso.org/standard/53820.html>
- [12]. *Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes — Safety Requirements for Portable Sealed Secondary Cells*, IEC Standard 62133-2, 2017. [Online]. Available: <https://webstore.iec.ch/publication/60990>
- [13]. *Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specification*, IEEE Standard 802.11, 2016. [Online]. Available: https://standards.ieee.org/standard/802_11-2016.html
- [14]. *Raspberry Pi 5 Product Brief*, Raspberry Pi Ltd., 2023. [Online]. Available: <https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-product-brief.pdf>
- [15]. *PCA9685 16-Channel, 12-Bit PWM Fm+ I²C-Bus LED Controller Data Sheet*, NXP Semiconductors, Rev. 4, April 2015. [Online]. Available: <https://www.nxp.com/docs/en/data-sheet/PCA9685.pdf>
- [16]. *DS3235-270 Servo Motor Data Sheet*, Microrobotics. [Online]. Available: https://github.com/microrobotics/DS3235-270/blob/master/DS3235-270_datasheet.pdf
- [17]. *XL4016 180 kHz 40 V 8 A Switching Current Buck DC/DC Converter Data Sheet*, XLSEMI. [Online]. Available: <https://www.xlsemi.com/datasheet/XL4016%20datasheet.pdf>
- [18]. *LM2596 SIMPLE SWITCHER Power Converter 150-kHz 3-A Step-Down Voltage Regulator Data Sheet*, Texas Instruments, Rev. May 2021. [Online]. Available: <https://www.ti.com/lit/ds/symlink/lm2596.pdf>
- [19]. J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 3rd ed., Pearson Education, New Jersey, USA, 2005.

APPENDICES

ANNEX - 1. IEEE Code of Ethics (English and Turkish)



IEEE Code of Ethics



IEEE Code of Ethics (English)

We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

1. To accept responsibility in making decisions consistent with the safety, health, and welfare of the public, and to disclose promptly factors that might endanger the public or the environment.
2. To avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist.
3. To be honest and realistic in stating claims or estimates based on available data.
4. To reject bribery in all its forms.
5. To improve the understanding of technology, its appropriate application, and potential consequences.
6. To maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience.
7. To seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others.
8. To treat all persons fairly and with respect, without discrimination based on race, religion, gender, disability, age, or national origin.
9. To avoid injuring others, their property, reputation, or employment by false or malicious action.
10. To assist colleagues and co-workers in their professional development and to support them in following this code of ethics.



IEEE Etik Kuralları



IEEE Etik Kuralları (Türkçe)

IEEE üyeleri olarak, teknolojilerimizin dünya genelinde yaşam kalitesi üzerindeki etkisinin bilincinde olarak ve mesleğimize, meslektaşlarımıza ve hizmet ettiğimiz topluma karşı kişisel sorumluluğumuzu kabul ederek, en yüksek etik ve profesyonel davranış ilkelerine bağlı kalmayı taahhüt ederiz:

1. Kamu güvenliği, sağlığı ve refahı ile uyumlu kararlar almaktan sorumlu olmak ve kamuya veya çevreye zarar verebilecek unsurları zamanında açıklamak.
2. Gerçek veya algılanan çıkar çatışmalarından mümkün olduğunca kaçınmak ve mevcut olduğunda ilgili tarafları bilgilendirmek.
3. Mevcut verilere dayalı olarak iddia ve tahminlerde dürüst ve gerçekçi olmak.
4. Her türlü rüşveti reddetmek.
5. Teknolojinin anlaşılmasını, uygun kullanımını ve olası sonuçlarını geliştirmek.
6. Teknik yeterliliği sürdürmek ve geliştirmek; yalnızca eğitim ve deneyimle yetkin olunan konularda görev üstlenmek.
7. Teknik çalışmalara yönelik dürüst eleştirilerde bulunmak, hataları kabul edip düzeltmek ve başkalarının katkılarını doğru şekilde belirtmek.
8. Irk, din, cinsiyet, engellilik, yaş veya milliyet ayrımı yapmaksızın tüm bireylere adil ve saygılı davranmak.
9. Yanlış veya kötü niyetli eylemlerle başkalarına, mallarına, itibarlarına veya işlerine zarar vermekten kaçınmak.
10. Meslektaşların profesyonel gelişimine destek olmak ve bu etik kurallara uymalarını teşvik etmek.

ANNEX - 2. Student Ethic Form



FACULTY OF ENGINEERING
ELECTRICAL & ELECTRONICS ENGINEERING

GRADUATION
PROJECT

DECLARATION

Ethical Behavior and Professional Honesty

Fair and accurate assessment are key factors that must be at the core of education. Any act of a student that would cause unfair advantage to himself/herself or another student is considered an academic dishonesty. It is also academic dishonesty to copy someone else's work and make it look like his/her own. This is called theft.

Allowing someone else to copy your work and make it look like his/her own is also fraud. It is both wrong and unethical to provide all kinds of unfair profits and advantages. It is also wrong to prepare the assignments, projects or laboratory reports of the other students. Doing this is not ethical, neither. Likewise, having his/her own homework, project or laboratory report being prepared and written by someone else is also a complete deception and immorality.

Using another student's answers, homework or reports without his/her knowledge is nothing but stealing that student's labor and this is called theft. It is the same thing to pass the course by cheating in exams.

We should not forget that those who cannot learn the truth when they are students continue these mistakes after graduation and become one of the characters we complain about in the society.

We expect our students and alumni to have ethical and humanly behaviors during both their professional business and social lives. For this, we want you to promise us that you will obey the ethical rules by signing this form.

Let us fight with unethical behavior together.

Prof. Dr. İsmail H. ALTAŞ
Department Chair

I believe what is written above is correct within the framework of ethical rules. I promise to behave ethically during and after my education.

Student No	Student Name and Surname	Date	Signature
210202009	Berna Meltem YILDIRIM	08/05/2026	
210202003	Ahmet Emirhan GÖKTÜRK	08/05/2026	
210202015	Sıla ÖZDEMİR...	08/05/2026	
210202023	Uygar BAŞ	08/05/2026	

Each student must fill out this form with a blue pen, sign it, and add it to Graduation Project file as an Appendix

Constraint Category	Explanations
Design Constraints	Robot body dimensions, structural stability, center of mass, DS3235 servo torque limits, and compact placement of Raspberry Pi, PCA9685, IMU, camera, TP-Link adapters, emergency stop button, and 12 V Li-ion battery are the main design constraints. I2C, CSI, USB interfaces and high-current servo operation require proper wiring, voltage regulation, and electrical safety.
Prototype Manufacturing Constraints	The prototype is manufactured using 3D printing. Strength limitations of ABS/PLA materials, layer adhesion, maximum part size of the printer, and tolerance limitations affect final part dimensions. Supply of electronic components, heat dissipation of control boards and EMI shielding are considered.
Implementation / Operational Constraints	In field deployment, the constraints include energy limits, maintenance frequency, servo overheating, wireless latency, and signal loss within disaster debris. Sudden power loss and risk of overcurrent requires safe shutdown procedures.
Social Impact Constraints	The robot contributes to search and rescue operations before human entry into hazardous areas; it should not, however, impede field teams, being mindful of the AFAD/municipality emergency workflow, nor create false victim signals that mislead rescue efforts.
Health Constraints	The system design shall not threaten personnel. Rotating joints and servos require mechanical shielding. Li-ion battery requires safe charging/discharging management to prevent explosion risk. Wireless communication shall not exceed the safe RF exposure limits for operators.
Environmental Constraints	The prototype shall prevent material leakages, heat damages, or environmental contaminations. Plastic parts, batteries, lubricants, and electronic wastes shall be disposed of following environmental regulations when the prototype's lifetime is complete.
Safety Constraints	Emergency stopping, short-circuit protection, safe cut-off of the battery, over-temperature protection, and proper grounding are very important. In printed parts, sharp edges must not be used. During operation, adequate safe approach distance must be maintained between the robot and operator.
Financial Constraints	Component selection was mostly constrained by budget. No industrial motors or sensors could be used and low-cost versions had to be picked instead, such as DS 3235 servos and a PCA9685 PWM module.
Legal Constraints	Use of wireless communication modules must comply with national radio frequency regulations. Sensor processing and images collected must comply with KVKK/GDPR personal data privacy laws. Field use requires permissions from emergency authorities.

ANNEX - 4. Interdisciplinary Work

Interdisciplinary Study	Explanations
Workshops or courses organized by the Head of the Department	During this project work, experiences were also gained in interdisciplinary workshops or courses organized by the Head of the Department and compulsory to attend. Throughout the semester, interdisciplinary seminars on engineering ethics, project management, and problem-solving methodologies were attended within the frame of compulsory interdisciplinary studies. These studies contributed to establishing communication across engineering disciplines and improved systematic decision-making and teamwork skills. Further, some studies for the Graduation Project necessitated collaboration outside the department. Within the framework of the needs related to the mechanical design of the quadruped robot, Volkan Yıldız, a Mechanical Engineering graduate from Çankaya University, was reached for support related to Solid modeling and the principles of mechanical design. Around 12 hours of consultancy were devoted to the work on CAD modeling, selection of mechanical reinforcement, and tolerance analysis for 3D-printed pieces. Additionally, the project team members attended online technical seminars, viewed Solid design tutorials, and engaged in video-based training resources to enhance interdisciplinary competencies. The out-of-department interactions greatly contributed to completing the robot chassis and leg structure according to the mechanical design constraints.

ANNEX - 5. Other Appendices

A5.1. Inverse Kinematics Software Code

This appendix presents the inverse kinematics function used for calculating the joint angle of a quadruped robot leg. The algorithm is based on a two-link leg model consisting of femur and tibia segments. The law of cosines is used to determine the joint angle required for a given target position in Cartesian coordinates.

Code Snippet 4.1. Inverse Kinematics Function

Function q = inverse_kinematics (x, y)

% L1 and L2 represent the leg segment lengths (in mm)

% L1: Femur length

% L2: Tibia length

L1 = 100;

L2 = 100;

% Inverse Kinematics Calculation (Result in Radians)

% This formula determines the required joint angle

% for a target (x, y) position

$$D = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2} \quad (4.1)$$

% Constraint check to prevent complex values

q = acos(max(min(D, 1), -1));

end

ANNEX - 6. Short Biographies

ANNEX 6.1. Berna Meltem YILDIRIM

Berna Meltem Yıldırım was born on 30 July 2002 in Ankara, Türkiye. She completed her primary, secondary, and high school education in Ankara. She is currently a fourth-year student in the Department of Electrical and Electronics Engineering at Ostim Technical University.

ANNEX 6.2. Sıla ÖZDEMİR

Sıla Özdemir was born on 26 July 2002 in Ankara, Türkiye. She completed her primary, secondary, and high school education in Ankara. She is currently a fourth-year student in the Department of Electrical and Electronics Engineering at Ostim Technical University.

ANNEX 6.3. Uygur BAŞ

Uygur Baş was born on 16 March 2002 in Ankara, Türkiye. He completed his primary, secondary, and high school education in Ankara. He is currently a fourth-year student in the Department of Electrical and Electronics Engineering at Ostim Technical University.

ANNEX 6.4. Ahmet Emirhan GÖKTÜRK

Ahmet Emirhan Göktürk was born on 01 March 2002 in Ankara, Türkiye. He completed his primary, secondary, and high school education in Ankara. He is currently a fourth-year student in the Department of Electrical and Electronics Engineering at Ostim Technical University.

ANNEX - 7. Check list of the submission requirements for Graduation Project II



FACULTY OF ENGINEERING
ELECTRICAL & ELECTRONICS ENGINEERING

GRADUATION
PROJECT – I
EVALUATION

GRADUATION PROJECT II - FILE SUBMISSION CHECKLIST

STUDENT INFORMATION

DATE: 08 / 05 / 2026

Number	Name and Last name	Signature
210202009	Berna Meltem YILDIRIM	
210202003	Ahmet Emirhan GÖKTÜRK	
210202015	Sıla ÖZDEMİR	
210202023	Uygar BAŞ	

This form will be filled out and signed according to the number of students in the project. Students who sign and submit this file accept the accuracy of the information they provide below.

PROJECT INFORMATION

Project Title	:	Design and Implementation of a Search and Rescue Robot Dog for Disaster Areas
Project Supervisor	:	Dr. Şenol GÜLGÖNÜL
	Signature	

FOR THE PROJECT FILE TO BE SUBMITTED

All of the following answers must be YES.

1. Is the number of verbatim quotations from any source in your dossier LESS than 20% and is the number of paragraphs of verbatim quotations zero? YES ☒ NO ☐
2. Does your dossier include a preface, abstract, main sections (Introduction, Theoretical Background, Design, Simulation, Experimental Study, Results, Evaluations) and a list of references? YES ☒ NO ☐
3. Is there an Introduction, Literature Review, Originality (novelty), Methodology, Impact, List of Standards and Work Schedule in the Introduction section? Is there leadership sharing and B plans in the work packages? YES ☒ NO ☐
4. Has a financial analysis commentary and a legal assessment that may arise from the project been added at the end of the design section? YES ☒ NO ☐
5. Are all equations numbered, all figures captioned and all charts captioned? YES ☒ NO ☐
6. Are all equations, figures, charts and references cited in the text? YES ☒ NO ☐
7. Are all the figures in the Design, Simulation and results section yours? YES ☒ NO ☐
8. Are all references listed in accordance with the Graduation Project Writing Guidelines and at least 5 scientific articles, theses and dissertations are cited? YES ☒ NO ☐
9. Have you included the IEEE Code of Ethics in the Appendix? YES ☒ NO ☐
10. Have you conducted interdisciplinary work and explained this in the Appendix? YES ☒ NO ☐

In order for the project file to be delivered,
all of the above answers must be YES.

ANNEX - 8. Graduation Project Student Tracking Form



FACULTY OF ENGINEERING
ELECTRICAL & ELECTRONICS ENGINEERING

GRADUATION
PROJECT

Student Number, Name and Surname : 1.210202009, Berna Meltem.YILDIRIM
: 2.210202003, Ahmet Emirhan GÖKTÜRK
: 3. 210202015, Sıla ÖZDEMİR
: 4. 210202023, Uygur BAŞ.

Project title : Design and Implementation of a Search and Rescue Robot Dog for Disaster Areas

Supervisor Title, Name and Surname : Dr. Şenol GÜLGÖNÜL

STUDENT WORK TRACKING SHEET

Education Year 2025 / 2026

Date visited	Status of the Graduation Project	Advisor's signature
Week 1	The mechanical design reviewed final dimensions determined	Sr
Week 2	Mechanical parts were produced using 3D Printer	Sr
Week 3	Electronic components sensors motor driver placed on the robot	Sr
Week 4	Motor control test	Sr
Week 5	Sensor reading and communication between controller was checked	Sr
Week 6	Walking algorithm was implemented	Sr
Week 7	Basic leg movement tests were performed	Sr
Week 8	Midterm Week	Sr
Week 10	Balance control algorithm was implemented PID parameters adjusted	Sr
Week 11	The robot tested on different surface	Sr
Week 12	Mechanical software and communication problems fixed	Sr
Week 13	Final communication balance and motion test were performed	Sr
Week 14	System was evaluated according to the project objectives	Sr
Week 15	Final report poster and presentation were prepared	Sr
Week 16	Final project documentation and presentation preparations were completed	Sr

Öğrencilerin veya çalışma gruplarının haftalık toplantıları sırasında bu formu proje yöneticisine imzalatmaları ve dönem sonunda proje raporuyla birlikte teslim etmek üzere saklamaları gerekmektedir.

ANNEX - 9. TUBITAK Project Closure Document

After the completion of projects such as 2209/A-B supported by TÜBİTAK, the signed version of the closure form that the advisor and students will receive from the TÜBİTAK project page should be added only to the Graduation Project as a final appendix.